

Retrieval-Augmented Generation-aided causal identification of aviation accidents: A large language model methodology

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ABSTRACT

Identification and assessment of key information within historical aviation accident reports are essential for preventing future accidents. However, manual causation extraction of key details is a labor-intensive and inefficient process due to the unstructured or semi-structured format of reports. To narrow these gaps, this paper proposes a Retrieval-Augmented Generation (RAG)-aided Causal Identification (RACI) model combining the Large Language Model approach. This novel model comprises a retrieval module and a generation module. Leveraging the real-world aviation reports published by the National Transportation Safety Board (NTSB), the proposed model demonstrates a substantial improvement in both retrieval accuracy and robustness. Specifically, the RACI method achieves a mean Average Precision (mAP) of 0.815 and a maximum mean Normalized Discounted Cumulative Gain (mNDCG) of 0.761. This study further examines the influence of various prompt designs on the reason localization performance of RACI. With the enhancement of RAG and noise filter method, RACI shows great robustness to noisy inputs. This highlights RACI's potential applicability in assisting the experts with causation identification. The experiment results demonstrate that RACI can serve as an efficient expert system to enhance decision making and risk management in aviation safety.

1. Introduction

In recent years, the passenger traffic in the aviation sector has experienced significant and rapid growth. According to the International Air Transportation Association (IATA), global air passenger traffic is expected to reach 7.8 billion by 2036, nearly doubling the 4 billion passengers recorded in 2017 (Aviator., 2017). Data from the International Civil Aviation Organization (ICAO) indicate that the aviation accident rate fluctuated between 2 % and 3 % per million passengers annually from 2017 to 2021 (ICAO, 2022). Given the catastrophic consequences of aviation accidents, there is an urgent need to ensure aviation safety. Historical accident cases serve as critical sources of information for identifying the underlying causes of accidents. Specifically, accident reports for aviation industry offer comprehensive analyses of the events preceding an accident, offering critical insights into potential risks and operational deficiencies from multiple perspectives. As aviation safety accident investigation reports are mainly composed of professional and unstructured texts, users need a strong professional background to manually identify key information (Yan et al., 2023). There is a lack of an automated and expertise-embedded

system for crew members and engineers that identifies causes quickly and precisely.

Automating the causal analysis of aviation accident texts using traditional methods, such as association rule algorithms (Nazeri and Bloedorn, 2004, Abesamis et al, 2020) and Bayesian networks (Zhang and Mahadevan, 2021), often requires significant manual intervention to construct causal logic. These methods are constrained by the quantity and quality of causal factors in the corpus, making them inflexible and unable to adapt to changing real-world scenarios. Traditional natural language processing (NLP) models faced similar challenges. These models were primarily used to classify accident texts or find similar incidents based on word frequency. However, they were highly specialized, with a focus of accuracy and generally lacked strong generalization capabilities for changing real-world scenarios (Shi et al, 2023).

In recent years, Language Models (LMs) have achieved remarkable results in various areas, such as chatbots, text generation, and text classification. Extracting information from natural language text was previously difficult for machines to handle, but LMs have made great strides in this regard. Pre-trained language models (PLMs) (e.g., BERT

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and GPT) have significantly transformed the field of NLP. For numerous language tasks, PLMs have achieved state-of-the-art performance (Min et al., 2023). In previous studies, Kiciman et al. (2023) illustrated that Large Language Model (LLMs) can make fact-based judgments based on causal relationships, making it feasible to use large language models for causal inference and accident cause identification. With the good performance of LMs, it is possible to use NLP technology in analyzing aviation safety event reports. However, the performance of PLMs also faces several challenges. PLMs are struggled in expanding or modifying their memory and may produce “hallucinations” (Marcus, 2020). To address these issues, Retrieval-Augmented Generation (RAG) has been proposed as a feasible solution (Zhou et al., 2024). RAG has demonstrated promising performance in facilitating efficient and accurate code generation (Gou et al., 2024, Parvez et al., 2021), chatbots (Zhou et al., 2024) and other applications. In general, a standard RAG process requires a retriever and a generator. The retriever is usually a lightweight model or a medium pre-trained language model, while the generator is typically a LLM. The RAG process involves retrieving relevant documents, combining them with the query as a prompt, and then generating the output using an LLM.

When investigating the causes of aviation accidents, retrieval models encounter several challenges, including limited input data, incomplete or fragmented information, lexical inaccuracies, and a high volume of irrelevant or extraneous data. In the RAG process, the accuracy of the retrieval process has a critical impact on the final generation results (Gou et al., 2024). However, each model has distinct shortcomings that hinder their performance in specific contexts. BERT heavily relies on pre-training on large corpora and often struggles with long-term dependencies, which can lead to less coherent results in generation tasks. GPT can generate contextually accurate responses, but may lack the precision and reliability needed in retrieval processes. Models mentioned before often overlook the importance of a well-structured retrieval process. Thus, the gap addressed in this research is the development of a more robust retrieval mechanism that enhances the performance of RAG process, ensuring that the generation result is more reliable and contextually grounded. This study proposes a novel RAG-Aided Causal Identification (RACI) model that includes a noise filtering module. RACI model consists of two main parts, which are Retriever and Generator. In the Retriever module, the model incorporates a filtering mechanism to enhance input quality. This filter is constructed using training labels derived from aviation accident reports, which contain causal statements represented by key terminology. The complete noise filtering method is combined with labeled dataset and adversarial training process. The filtering process enhances both the quality of input and the noise-resistance ability of model. This is in contrast to generic noise filters that may not be sensitive to domain-specific nuances. By matching potential combinations of these causal statements, the filter ensures more precise and relevant information retrieval. The main contributions of this study are as below:

- **Pioneering the use of Large Language Models for cause identification in aviation safety:** This study is the first to propose the application of Large Language Models to identify causes in aviation accident reports.
- **RAG method for addressing short and insufficient input in cause identification:** The proposed model in this study employs RAG methods to tackle the issues of short and insufficient input information during the process of cause identification.
- **Noise filtering method for input texts to enhance retriever performance:** This study proposes a noise filtering method for input texts, which enhances the performance of the retriever by improving the accuracy of the information retrieval process.
- **Real-time decision support for aviation safety:** The RACI model provides a dynamic tool for aviation safety experts, offering real-time causal summaries of past similar aviation accidents. These insights

are expected to support cabin crews and engineers by facilitating informed decision-making.

In summary, the novelty of this study lies in its integration of cutting-edge language models, RAG methods, and noise filtering techniques to improve the identification of causal factors in aviation safety. The RACI model not only addresses key limitations in current approaches but also paves the way for more reliable and actionable insights in real-time aviation operations.

2. Literature review

Accurate causal analysis of aviation accidents requires a deep understanding of the methodologies that have evolved over time. The literature in this domain spans traditional statistical models, rule-based approaches, and more recently, advancements in Natural Language Processing (NLP). These developments highlight the progression of techniques used to analyze complex and unstructured data. The following subsections review these methodologies, beginning with traditional approaches and transitioning into modern NLP-based methods. This study also identifies the gaps in these approaches that the proposed Retrieval-Augmented Generation (RAG)-aided Causal Identification (RACI) model aims to address.

2.1. Traditional methods in aviation accident analysis

Mining causal relationships from unstructured text data is a challenging task (Liu et al., 2018). The primary research methods for early aviation attribution include association rule algorithms, Bayesian algorithms and traditional expert systems. Nazeri and Bloedorn (2004) applied association rule algorithms to aviation safety incident reports and demonstrated the effectiveness of this method through experiments. Zhang and Mahadevan (2019) constructed Bayesian networks by devising a four-step process, using visualization methods to represent aviation accidents as graphs. This approach captures causal relationships within the accident sequence, focusing on aircraft damage and personnel injuries, and calculates prior and conditional probabilities. Ng et al. (2022) proposed an expert system with fuzzy logic method to support aviation occurrence investigations. The primary algorithms mentioned above need substantial attention from aviation domain experts in their life cycle. While effective in capturing relationships between variables, these models often struggled with handling the vast amounts of unstructured data present in accident reports and required significant domain expertise for setup and interpretation. These early techniques laid the foundation for causal analysis but were constrained by their reliance on expert input and limited capacity to process large datasets efficiently.

2.2. NLP in accident attribution

As the volume of unstructured data in aviation safety reports grew, researchers turned to more automated techniques, particularly in the field of Natural Language Processing (NLP). Tanguy et al. (2016) studied the performance of Support Vector Machines (SVM) in classifying aviation accident reports, finding that SVM could achieve over 95 % accuracy for certain event classifications, though its prediction stability was relatively poor. Zhang and Mahadevan. (2019) constructed an integrated model of SVM and deep learning to extract features from aviation accident reports, categorizing risk levels into high risk, medium-high risk, medium risk, medium-low risk, and low risk to classify aviation accidents. Perboli et al. (2021) used a similarity calculation method based on the SHEL standard accident causation model for text classification to identify human factors in aviation accidents, and tested this on a set of real accident data. Liu and Yang. (2022) developed a railway hazard identification model with LSTM method. Combined with knowledge graph, that approach realized the visualization and

quantitative description of the potential relationship among hazards, faults, and accidents. Bharathi et al. (2024) developed a model combining rule-based and supervised methods to propose a named entity recognition system in the aviation field. Gao et al. (2024) proposed an effective semantic encoding algorithm to enhance the representation of aviation terms in word embeddings by creating a text similarity dataset. Zhou et al. (2024) proposed a risk analysis model that integrates a temporal convolutional network (TCN) and a dual attention gated recurrent unit (DAGRU) neural network. The proposed TCN-DAGRU model showed the best prediction accuracy among seven deep learning models. Tikayat Ray et al. (2023) discussed the advantages and disadvantages of generative large language models in the context of aviation safety and proposed a human-in-the-loop system to ensure responsible and effective use of such models.

A primary research focus is extracting causal factors from the text of these reports. For example, Ni and Zhang (2023) employed NLP techniques to extract, classify, and structure the causes of accidents from 137 Chinese road transportation reports. They then used the structured data to construct a Bayesian network, enabling the identification of risk factors from the accident reports. Jiang and Wang (2024) proposed a Transformer-based causality extraction model for railroad accidents, which was combined with a BiGRU (Bidirectional Gated Recurrent Unit) to enhance the location information within the Transformer architecture. The final model outperformed the best existing model at that time. To address the high computational cost associated with retraining the BERT model for accident report analysis, Chen et al. (2022) integrated a convolutional neural network (CNN) with the BERT model, resulting in an efficient accident causation classification algorithm. Additionally, using NLP techniques to obtain information from outside traditional traffic scenarios can also help in assessing traffic conditions. For instance, Chang et al. (2022) developed a framework to identify accident- and congestion-prone areas that are widely discussed on social media. Their approach determines accidents and congestion by analyzing the emotional content of relevant social media posts. While these methods improved accuracy in identifying human factors and specific incident details, they often lacked robustness in the presence of noisy data or incomplete reports.

2.3. Large language models (LLMs) in aviation and traffic accident analysis

The recent advancements in large language models (LLMs), such as GPT and BERT, have significantly influenced the field of NLP, especially in tasks requiring deeper understanding and generation of text. As a result, integrating LLMs with traffic accident analysis has become a popular research direction. de Zarzà et al. (2023) explored the role of multimodal data in predicting traffic accidents, utilizing LLMs alongside Visual Language Models (VLMs) to process the multimodal information. Chen et al. (2024) proposed a Claude-prompt-based method for extracting causality information from aviation accident reports, which significantly enhances the accuracy of information extraction. Similarly, Jaradat et al. (2024) employed LLMs to extract valuable insights from social media data for real-time road traffic safety analysis. However, large models suffer from dependence on training data and poor understanding of information in specialized domains (Cuconasu et al., 2024).

RAG methods are widely used in natural language text processing (Guu et al., 2020, Xu et al., 2021, Zhao et al., 2022, Min et al., 2021, Lewis et al., 2020). The RAG model (Guu et al., 2020) and the REALM mode (Lewis et al., 2020) introduce the retrieval process into a retrieval-enabled paragraph-indexed corpus for augmented conditional generation. This process uses the corpus as a source of knowledge, and this extension greatly expands the information available to the model, resulting in significant computational cost savings. In traffic accident analysis, the lack of open-source large models pre-trained specifically for traffic scenarios makes it a viable research solution to use RAG technology. RAG can expand the model's knowledge base, allowing large

models to handle traffic scenarios with greater accuracy. In this study, the goal of the RAG is to quickly retrieve similar safety incidents based on the input, and extract the key causes from the similar safety incidents. The information from similar accidents can help professionals deal with the current accident. This progress can be regarded as a special Question-Answering (QA) problem.

With the help of RAG, LLMs can answer questions about private data or previously unseen documents. Guu et al. (2020) used the RAG approach to augment pre-training on open-domain Q&A questions and achieved excellent performance. Ram et al. (2023) proposed a retrieval-augmented In-Context Large Language model architecture. With pre-positioning the context in the input, this architecture avoids further changes to the model architecture. At the same time, model still show impressive gains. Wu and Ma. (2024) developed a retrieval framework to mine requirements and requirement-related documents of construction. The system can satisfy 91 % of the project managers' requirements. Jiang et al. (2023) proposed Forward-Looking Active Retrieval augmented generation (FLARE) method, which uses the predicted next sentence as the basis for retrieval, and the FLARE method has achieved competitive results on several tasks. Peng et al. (2023) proposed an LLM-AUGMENTER system, the system is plug-and-play when combined with LLMs. It can be used to extend the database content and enhance the performance of LLMs. With the help of this system, ChatGPT's hallucinations significantly reduced. And system does not affect the smoothness of the answer. Xu et al. (2024) combined the RAG with knowledge graph to assist in Q&A, and the model ultimately reduced the median problem-solving time by about 26 % in real-world scenarios. Siriwardhana et al. (2023) evaluated the impact of co-training of the Retriever and Generator on the task of ODQA, and proposed the RAG- end2end method. The co-training method realized the Q&A task to be domain-adapted. Samarajeewa et al. (2024) introduced a graph-based causal extraction method augmented with RAG method. This research proves the objective influence of RAG method on causal extraction. However, the research doesn't fully utilize the real-time Q&A capabilities of LLMs.

In summary, NLP techniques have been widely used in traffic accident analysis. The recent advancements in LLMs have provided a significant boost to the analysis of tawaraffic accident-related texts. However, LLMs face critical and practical challenges, such as limited database and inadequate understanding of specialized knowledge. In the application of large language models in aviation safety, issues and challenges include inaccurate interpretation of aviation-specific terminology and frequent lack of context information in input data (Yang and Huang, 2023).

2.4. Research Gaps

While existing research has made considerable progress in automating causal analysis in aviation and traffic accidents, significant gaps remain. Early statistical and rule-based models, while effective in their time, struggle to handle large, unstructured datasets and require substantial manual effort. More recent NLP approaches, particularly those using large language models, show promise but often fail to fully capture the complexity of aviation-specific data, suffering from issues like poor noise resistance and limited domain understanding. Additionally, existing models often lack an effective retrieval mechanism, which is crucial for ensuring the accuracy and relevance of the generated insights, especially when dealing with complex or ambiguous accident reports.

The proposed Retrieval-Augmented Generation (RAG)-aided Causal Identification (RACI) model addresses these gaps by integrating a retrieval mechanism with advanced NLP techniques. The RACI model leverages a pre-trained Sentence-BERT (SBERT) for the retrieval module, enhanced by adversarial training to improve robustness against noise. By integrating this with the Qwen2-7b language model for causal generation, the RACI model offers an efficient, scalable solution for extracting accurate causal factors from aviation accident reports. This

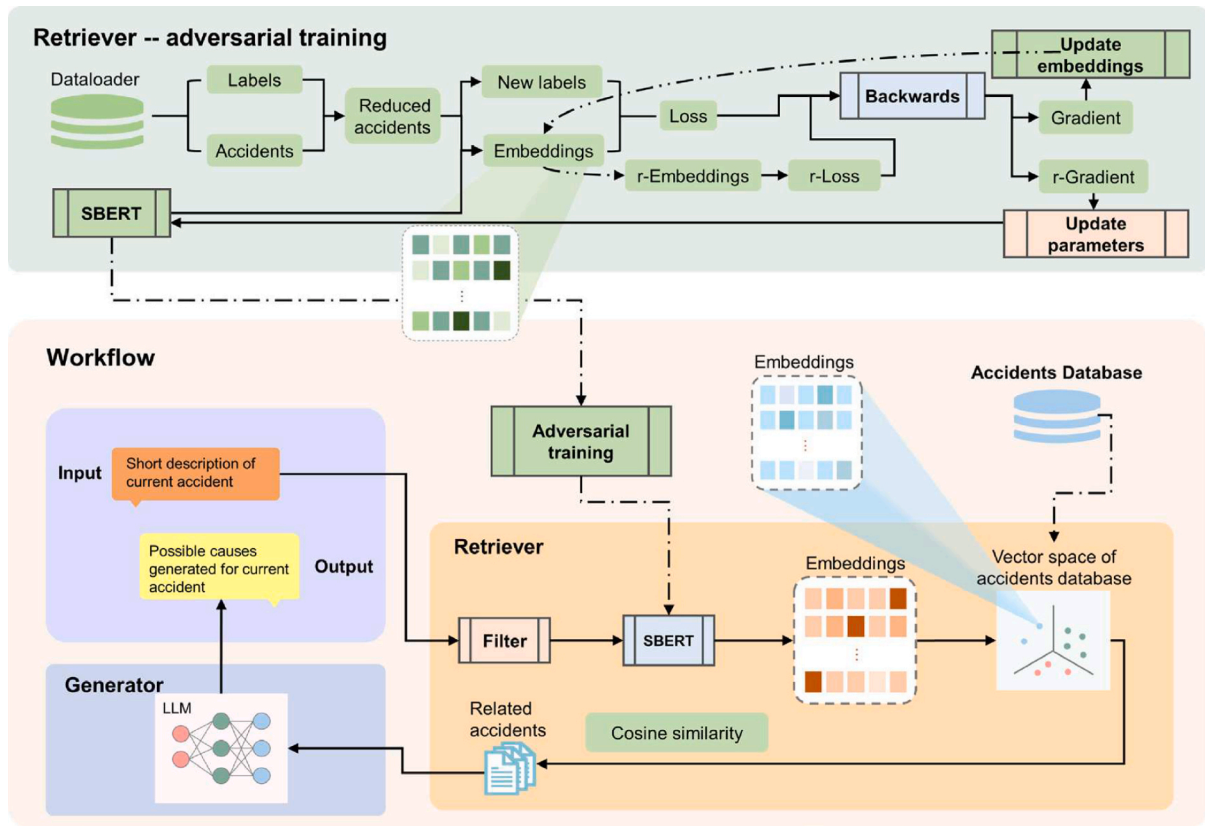


Fig. 1. Framework of RACI model.

model significantly improves retrieval accuracy and robustness compared to existing systems, providing a more reliable tool for aviation safety professionals.

3. Method

This section is dedicated to discussion about the development of the RACI model and the evaluation parameters to assess the model performance. This section explains the components of RACI including Retriever module and Generator module. The framework of RACI model is shown in Fig. 1.

3.1. Framework of RACI

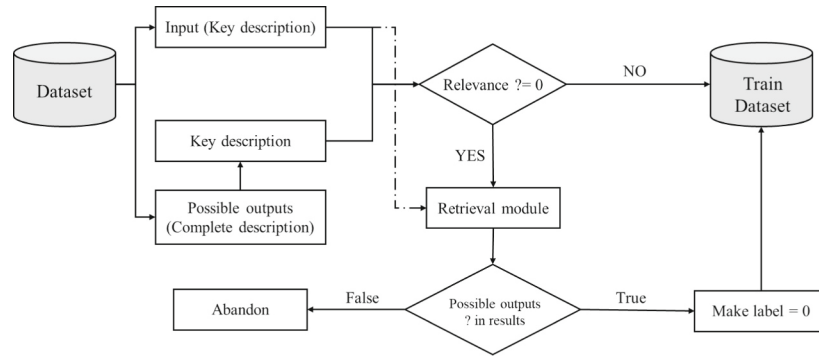
The RACI model comprises two primary components: the **Retriever** and the **Generator**. The **Retriever** operates in two distinct stages. In the first stage, it processes and filters the input text, transforming it into concise segments that align with the model's retrieval criteria. This operation enhances the efficiency of information extraction. In the second stage, Retriever uses the converted short text to retrieve relevant accidents from a large accident text database. These retrieved accidents are used as the basis of further text generation. The retrieval process is accomplished through the SBERT model. In order to improve the anti-noise ability of the retrieval module, the study conducts adversarial training on SBERT model. In **Generator**, the retrieved text is concatenated and combined with other parts of the prompt as input to Generator.

3.2. Data reduction and retrieval model training with adversarial training method

3.2.1. Data reduction

As retrieval process can be seen as a semantic similarity comparison task. The supervised learning process of retrieval module needs paired examples to serve as training dataset. However, the total number of possible combinations of input and output will be too large and significantly reduce the training efficiency. The reduction of dataset size is needed. In once retrieval, the accurate outputs usually are a small part of total possible outputs. That means most of the similarity scores of the input and possible outputs should be 0. If we simply consider every possible combination, the training dataset will be highly sparse. We can reduce the amount of 0s in dataset to make dataset smaller without losing too much valuable information.

Take the data from the National Transportation Safety Board (NTSB) as an example. The possible inputs (regards to the key description of an accident) contain 683 mutually exclusive data points, while the possible outputs (regards to the complete record of an accident) contain 1,797 mutually exclusive data points. If we consider every possible combination, this results in a dataset D with 1,227,351 data entries (the product of 1,797 and 683). Considering the text length and the dataset's size, directly using D for training would lead to extremely long training times. Additionally, it was observed that the labels in D are highly sparse, with the majority of the similarity labels being 0. Pre-trained models already possess the ability to distinguish between highly dissimilar texts, where the relevance of two distinct texts is expected to be zero. This capability provides an opportunity for further optimization. Based on these assumptions, the study applies the following preprocessing steps to the



Notes: The single-dotted line indicates directly passing the input to the retrieval module.

Fig. 2. Process of labeling database.

original dataset: In the first step, only the subset of D with non-zero similarity labels is generated. To correct cases where the pre-trained model incorrectly predicts a non-zero relevance for data that should have a relevance score of 0, the study uses a relevance score calculation algorithm based on inclusion relationships. If the key description corresponding to possible output has a score of 0 with key description of input in the inclusion relationship-based similarity algorithm, it indicates that the model has incorrectly judged the correlation between input and possible output. In such cases, a new statement pair is created using possible output and input, and the label is set to 0 before adding it to the training dataset. The detailed processing method is illustrated in Fig. 2.

By processing the data as shown in Fig. 2, the study reduced the training dataset size from 1,227,351 entries to 188,882 entries, resulting in a significant reduction in the dataset size.

3.2.2. Adversarial training

In terms of model development, this study incorporates Adversarial Training (AdvT) to enhance model performance. AdvT is widely used as a training method in the research of image neural networks. However, because text inputs are discrete rather than continuous, AdvT cannot be directly applied to NLP tasks. To address this, Miyato et al. (2016) proposed an adaptation that applies adversarial training in a continuous word embedding space, leading to improved model robustness and better performance across various NLP tasks.

In this study, AdvT is implemented by perturbing the word vector layer with Fast Gradient Method (FGM). FGM process for adversarial training includes:

1 Compute the gradient of the loss function: Given an input embedding x , the model computes the loss $\mathcal{L}$, which measures the difference between the predicted and expected outcomes (e.g., cosine similarity for retrieval tasks). The gradient of $\mathcal{L}$ with respect to the input embedding is calculated as

$$\nabla_x \mathcal{L} = \frac{\partial \mathcal{L}}{\partial x} \quad (1)$$

2 Generate adversarial perturbation: The adversarial perturbation δ_x is calculated as

$$\delta_x = \epsilon \cdot \frac{\nabla_x \mathcal{L}}{|\nabla_x \mathcal{L}|_2} \quad (2)$$

where ϵ is a small hyperparameter controlling the magnitude of the perturbation, and $|\nabla_x \mathcal{L}|_2$ is the l_2 -norm of the gradient.

3 Apply the perturbation: The adversarial example x_{adv} is created by adding δ_x to the original input embedding x

$$x_{adv} = x + \delta_x \quad (3)$$

4 Update the model with adversarial loss: The perturbed embedding

x_{adv} is passed through the SBERT model to compute the adversarial loss $\mathcal{L}_{adv}$. The model parameters are then updated to minimize the combined loss

$$\mathcal{L}_{total} = \mathcal{L} + \lambda \cdot \mathcal{L}_{adv} \quad (4)$$

where λ controls the weight of the adversarial loss.

The calculation of adversarial perturbations requires gradient computations at the embedding level, leading to increased training time and memory usage. To address this challenge, this study reduces batch sizes to lower memory consumption. By using FGM-based adversarial training, the model is exposed to both the original and adversarial versions of the word vectors during training. This process enhances SBERT's ability to handle real-world noisy text, making it more resilient in retrieval tasks.

3.3. Retriever

The retrieval task can be seen as matching the input query with the target texts. To take it further, it refers to the assessment of the similarity between the query and documents within the database. This paper focuses on the semantic similarity between the input text and the accident text in the accident document database. Reimers (2019) proposed the Sentence-BERT (SBERT) model, which presents a network structure of twin tertiary levels to obtain fixed-length sentence vectors capable of interpreting the semantics, and thus utilizes cosine similarity to compute the similarity between the sentence vectors. Compared with the BERT model, the SBERT model accelerates the speed of similarity comparison while ensuring the accuracy of comparison. Therefore, SBERT model is chosen as the base model for RACI's retrieval stage. Building on the SBERT architecture, the Retriever module adds a filter to the front-end of the model and supervised training process to better align with the specific requirements of the research context.

3.3.1. Noise filter for input text

The development of setting filters is motivated by two considerations. First, in real-world scenarios, input text may contain noise such as spelling errors, sequencing issues, or missing words, which can significantly impact query accuracy. Filters should mitigate the impact of the aforementioned extraneous noises, thereby enhancing the RACI model's performance and robustness. Secondly, the inputs to the RACI model may vary greatly in sequence and words, posing challenges for the fine-tuned model to ensure accurate matching between these brief text segments and the corresponding event descriptions. The filter helps re-organize noisy input into a standardized short text.

The training data contains long text for the detailed description of the accident and short text for the flight stage and accident types. The correspondence between the short text and the long text is the basis of retrieval. At the same time, to addressing the limitations in effective

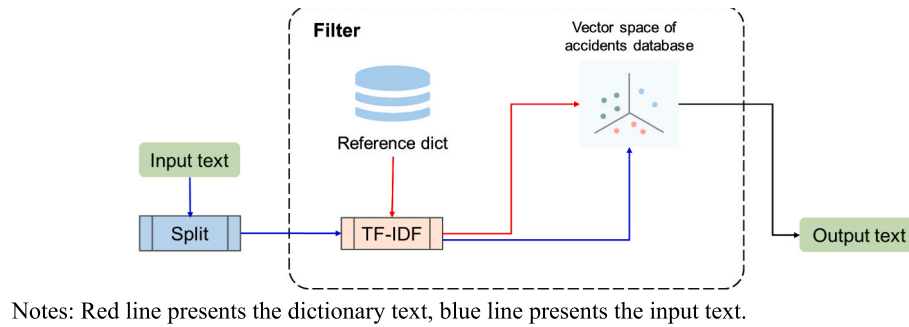


Fig. 3. Framework of filter.

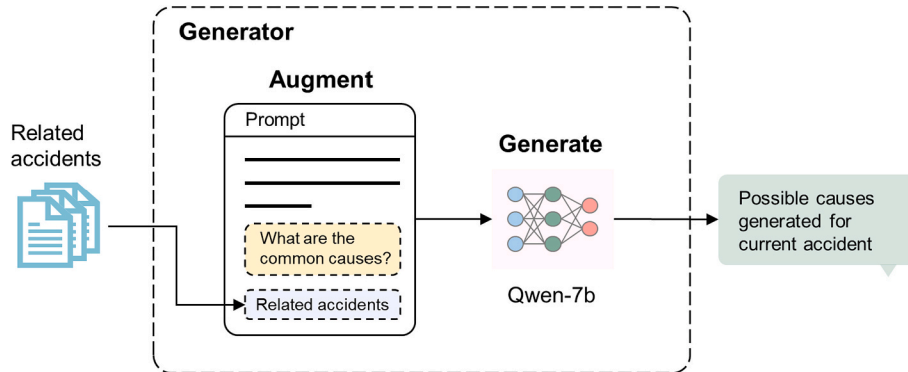


Fig. 4. Framework of Generator with Qwen-7b (Nazeri and Eric, 2004).

combination of short texts, this study constructed a specialized dictionary tailored for short-text processing. Using that dictionary, this study implements the filter for the input text. The specific filtering process is shown in Fig. 3. The filtering process employs the term frequency-inverse document frequency (TF-IDF) method to compare the cosine similarity between the input text and the dictionary, producing the reorganized output text. The TF-IDF method offers the advantage of requiring no prior training, enabling it to efficiently complete the filtering process with minimal impact on the model's running time. Since the filtering module relies on a domain-specific keyword-matching approach, maintaining high precision without excessive false negatives proved to be a trade-off that required multiple iterations of optimization.

3.4. Generation of causes

When localizing the cause during an accident, the inputs to RACI model are generally short and lack sufficiently informative information. It is impractical to use the information directly for model generation to find the probable cause of accident. Therefore, the study adopts RAG approach to assist the LLM for text generation by obtaining more similar incidents through the retrieval process. As the framework of Generator shown in Fig. 4, this study used the Qwen2-7b large model (Yang et al., 2024) for the text generation task. Qwen2-7b was developed by Alibaba, which is open source and easily accessible and has achieved excellent performance in text generation, code, and math.

The model maps the accident database and inputs into the vector space to match multiple related accidents strongly associated with the current one. The inputs obtained by the model are short texts and may contain interfering information, making them unsuitable as the query in the prompt. Instead, we fixed the query statement as "What are the common causes of the accidents," using the accident descriptions as the context. This ensures that the output is generated in the direction of cause-relatedness and utilizes the input text to achieve retrieval enhancement.

In summary, the RACI model constructs a Retriever and Generator module based on the labeled accident text of the NTSB, and integrates the past accident cases into a knowledge base to provide guidance for new accidents. It is noteworthy that although the RACI model is established through aviation accident data, its workflow does not rely on the specificity of aviation data. As long as an initial dataset of text data with causal labels is available to complete the training process of the Retriever, the RACI model has the potential to be applicable in more areas.

3.5. Evaluation parameters

In general, the evaluation results is affected by two primary aspects: the reasonableness of the manual scoring in the dataset, and the applicability of the evaluation indicators to the task objectives. Given that varying contexts impose distinct requirements on text similarity, it is typically essential to employ multiple evaluation metrics to achieve a comprehensive assessment that aligns with diverse task demands. Reimers et al. (2016) pointed out that when selecting text similarity evaluation metrics, it is critical to understand the task attributes and map them to the desired evaluation metrics.

According to the framework proposed by Beyer (2015), the text similarity task can be described in terms of three dimensions, which are mapping relations, clusters, and information. The mapping relation represents the ratio between the number of query documents and output documents and is categorized as 1 : 1, 1 : n, and m : n. Clustering describes the range of elements in the result set to be evaluated, typically categorized as all elements, the top-k elements, or elements meeting a specific threshold. Information represents the type of information in the result set, which is categorized into score value, ranking, and category. For instance, comparisons based on the magnitude of result scores fall under 'score value,' while assessments of ranking order, such as determining result similarity, are classified under 'ranking.' Simple categorization cases are defined as "categories".

Table 1

Evaluation parameters of model.

Task attributes	Evaluation parameters	Physical meaning
(1 : n, top-k, categories)	Average Precision (AP)	AP quantifies the system's ability to rank relevant causal factors for a given query higher than irrelevant ones.
(1 : n, top-k, ranking)	Normalized Discounted Cumulative Gain (NDCG)	mAP represents the average effectiveness of Retriever in ranking relevant causal factors across all aviation accident queries.
(m : n, top k, categories)	Mean Average Precision (mAP)	NDCG evaluates whether the most relevant causal factors are retrieved early in the ranked list.
(m : n, top k, ranking)	Mean Normalized Discounted Cumulative Gain (mNDCG)	mNDCG reflects the model's overall ability to prioritize highly relevant causal factors across all accident queries.

In the evaluation framework proposed by Reimers et al. (2016), the evaluation metrics for tasks with a mapping relation of 1 : n include average accuracy and normalized discounted cumulative gain (NDCG) (Ni and Zhang, 2023). For tasks with a mapping relation of m : n, in this paper, the mean value of m query results is used as the evaluation metrics. Different task attributes and their corresponding evaluation metrics are shown in Table 1.

a) Average Precision, AP.

When the model produces multiple results, and the total number of output results is N , arrange all the results in descending order based on the model's rating. Then, evaluate the precision of the top K results that the model identifies as similar. Finally, average the precision values to obtain the overall average precision.

$$AP = \frac{\sum_{k=1}^K Precision_k}{K} \quad (5)$$

$$Precision_k = \frac{TP_k}{k} \quad (6)$$

In Equation (1), $Precision_k$ is the precision of the top k outputs, K is the total number of results judged to be similar in the model output. In Equation (2), TP_k is the number of results from the top k outputs that are judged to be similar and are generally similar as well.

In cases where multiple query statements are involved, the average precision for each query can be calculated and then averaged to determine the mean average precision (mAP) of the results. The Equation is shown in 3.

$$mAP = \frac{\sum_{m=1}^M AP_m}{M} \quad (7)$$

AP_m is the average precision of the m th query statement and M is the total number of query statements.

b) Normalized Discounted Cumulative Gain, NDCG.

Normalized Discounted Cumulative Gain (NDCG) is a widely used evaluation method in retrieval and recommendation systems. It is based on Discounted Cumulative Gain (DCG) and Ideal Discounted Cumulative Gain (IDCG). DCG has two main principles: (1) highly relevant results have a greater impact on the score than results of lower relevance, and (2) the position of results with different relevance levels significantly influences the score. The formula for DCG is shown in Equation (4).

$$DCG = \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log_2(i+1)} \quad (8)$$

rel_i is a manual score for the i th outcome, p is the total number of output results, $1/\log_2(i+1)$ is a discount factor for the i th outcome.

IDCG is the value of DCG in the most ideal case, and its equation is shown in Equation (5).

$$IDCG_p = \sum_{i=1}^{|REL|} \frac{2^{rel_i} - 1}{\log_2(i+1)} \quad (9)$$

$|REL|$ denotes the set consisting of all the outputs sorted in descending order of relevance, taking the first p results.

NDCG is the normalized version of DCG, as comparing raw DCG values is not meaningful when the number of results returned by different query statements varies. Therefore, it is necessary to normalize the DCG value to a relative scale to enable comparison. The formula for calculating NDCG is shown in Equation (6).

$$NDCG_p = \frac{DCG_p}{IDCG_p} \quad (10)$$

When multiple query statements are involved, the NDCG values from all the query results can be averaged to obtain the mean Normalized Discounted Cumulative Gain (mNDCG). This is calculated using the formula shown in Equation (7).

$$mNDCG = \frac{\sum_{m=1}^M NDCG_m}{M} \quad (11)$$

$NDCG_m$ is the average precision of the m th query statement and M is the total number of query statements.

Table 2

Merged Data Format.

ev_id	narr_accf	narr_cause	occ_No	occurrence	occurrence_sum
20001206X01148	When landing gear extension was attempted in preparation for landing, the left main landing gear did not extend. The pilot made a go-around, completed the required emergency procedures, could not lower the left main landing gear, and diverted to another airport. The pilot landed with the left main landing gear in the retracted position. The airplane sustained minor damage. The 9 crewmembers and 141 passengers were not injured. Examination of the left main landing gear disclosed a broken landing gear door actuator fitting. Several service bulletins had been issued outlining repetitive inspections procedures aimed at preventing such failures. The operator had complied with all applicable service bulletins and airworthiness directives.	The total failure of the left main landing gear door actuator fitting due to fatigue cracking in the fitting. Contributing to the accident was inadequate written inspection procedures by the aircraft manufacturer.	1 2 3	Approach VFR pattern final approach Airframe / component / system failure / malfunction Descent –emergency Forced landing Landing –flare / touchdown Gear not extended	Approach VFR pattern final approach Airframe / component /system failure / malfunction, Descent –emergency Forced landing, Landing –flare / touchdown Gear not extended

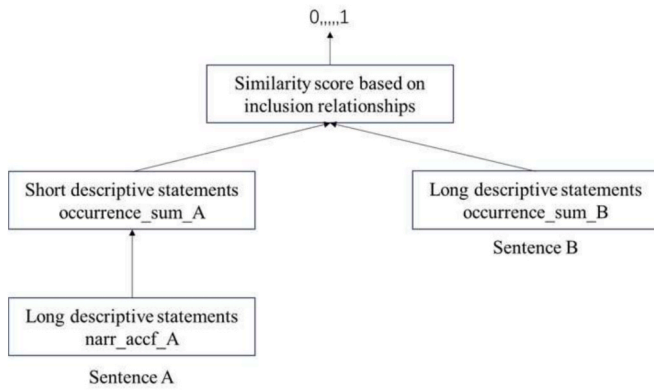


Fig. 5. Similarity score calculation by SBERT (Reimers, 2019).

Table 3

Occurrence_code inclusion hierarchy (partial).

First level	Second level	Label
Standing	Standing	500
	Standing – pre-flight	501
	Standing – starting engine(s)	502
	Standing – engine(s) operating	503
	Standing – engine(s) not operating	504
	Standing – idling rotors	505

Table 4

Phase_of_flight inclusion hierarchy (partial).

First level	Second level	Label
Forced landing	Forced landing	180
	Ditching	150
Fire/explosion	Fire/explosion	170
	Fire	171
	Explosion	172

4. Experiments and results

In this section, the process of training Retriever is discussed in detail, including the description of dataset used for training, the labeling method and evaluation.

4.1. Retriever training

This paper utilized aviation accident data from the National Transportation Safety Board (NTSB). This dataset contains aviation unsafe incident reports from 1982 to the present, with corresponding data extracts in Microsoft Access database format. The complete contents of the database can be accessed at <https://app.nts.gov/avdata>.

The relevant data tables include: “events”, “aircraft”, “narratives”, and “occurrences”. The “events” table contains basic information about each accident. Key data fields include “ev_id”, the unique code for NTSB aviation safety accident reports, and “Aircraft_Key”. The “Aircraft_Key” is used to distinguish multiple aircraft involved in a single accident. The “narratives” table provides detailed descriptions of the accidents. In this table, “narr_accf” represents an excerpt from the accident narrative, detailing the incident, while “narr_cause” provides a textual summary of the findings on the accident’s cause. The “occurrences” table documents the sequence of events in coded format. Each aircraft involved in an occurrence has separate entries in the occurrences table. The “occurrences_No” field represents the sequence of occurrences, with most event records containing multiple occurrences. The “Occurrence_Code” denotes the type of occurrence, and the “Phase_of_Flight” field indicates the flight phase during which the occurrence took place. Each

occurrence is described by combining the flight phase and occurrence type, forming the field occurrence. The complete description of the incident, represented by “occurrence_sum”, is generated by concatenating all occurrences in the sequence of their occurrence.

The NTSB dataset is highly organized and includes a well-defined short description field (occurrence_sum) that serves as a concise summary of each accident. This structure enables the development of an effective noise filter in the Retriever.

The object of this paper is to assess the aviation accidents of commercial operation, then 2,461 reports were selected with “far_part” of 121 and 129. Among these reports, there are 1,797 reports are adopted for this paper whose data are not empty. After data manipulation, the example accident in the final csv dataset format is shown in Table 2.

4.1.1. Similarity score labeling based on inclusion relationships

The effectiveness of fine-tuning depends on the structure and content of the text in the training set. In this paper, the NTSB data was selected as the training dataset. Specifically, the detailed accident description field (“narr_accf”), the causal description field (“Occurrence_code”), and the description of the flight phase at the time of the accident (“Phase_of_flight”) from the NTSB data were utilized.

A supervised training approach for the SBERT model during the fine-tuning phase is more likely to yield better performance and greater model stability. This study proposes a method for calculating similarity scores based on inclusion relationships to label the training data. The labeling process is based on the premise that both the “Occurrence_code” and “Phase_of_flight” fields consist of a finite set of numbered codes. These two fields are combined to form the Occurrence field, and multiple occurrences from a single incident are combined into the “occurrence_sum”. For any narr_accf_A associated with occurrence_sum_B, the study first retrieves the occurrence_sum_A corresponding to narr_accf_A and then calculates a correlation score between occurrence_sum_A and occurrence_sum_B based on the inclusion relationship. The calculation of this correlation score is illustrated in Fig. 5. The similarity score is obtained by SBERT (Reimers, 2019).

An “occurrence_sum” in the training data is generally formed by concatenating multiple occurrences, each being a combination of “Occurrence_Code” and Phase_of_flight. For any two occurrence_sum entries, such as occurrence_sum_A and occurrence_sum_B, the inclusion relationship operates on two levels: (a) the inclusion relationship between “occurrence_sum” entries, and (b) the inclusion relationship between occurrence.

(a) Inclusion relationship between occurrence

The sufficient condition for an inclusion relationship between two occurrences is that both the “Occurrence_code” and “Phase_of_flight” have an inclusion relationship. Based on the definitions and codes provided in the NTSB’s Aviation Coding Manual, the inclusion hierarchy of “Occurrence_code” and “Phase_of_flight” is established. The inclusion hierarchy is shown in Table 3 and Table 4. The first level in the table represents the major category, while the second level represents the minor category. According to the coding manual, major events belong exclusively to the major category, whereas minor events can be attributed to either the major or minor category. For this hierarchical structure, the study developed a judgment mechanism: if the input is a major category statement and the output matches a minor category event, the match is considered correct. However, if the input is a minor category statement and the output corresponds to a major category statement, the match is considered to be incorrect.

(b) Inclusion relationship between “occurrence_sum”

“occurrence_sum” is equivalent to “narr_accf” and is used to describe the complete sequence of events in an accident. “occurrence_sum” is created by concatenating multiple occurrences, and the inclusion

Table 5

Pseudo-code for similarity score calculation.

Algorithm Text correlation score based on inclusion relation
Input: input, output Output: score 1: score $\leftarrow$ 0; // The similarity score of input and output 2: for each input(phase, occ) $\in$ input do 3: num $\leftarrow$ 0; // The intersection number of input(phase, OCC) and output, 0 or 1 4: for each output(PHASE;OCC) $\in$ output do 5: if phase $\in$ p.keys and occ $\in$ o.keys and PHASE $\in$ p[phase] and OCC $\in$ o[occ] then 6: num $\leftarrow$ 1; 7: else if phase $\in$ p.keys and occ $\notin$ o.keys and PHASE $\in$ p[phase] and OCC = occ then 8: num $\leftarrow$ 1; 9: else if phase $\notin$ p.keys and occ $\in$ o.keys and PHASE = phase and OCC $\in$ o[occ] then 10: num $\leftarrow$ 1; 11: else if PHASE = phase and OCC = occ then 12: num $\leftarrow$ 1; 13: end if 14: end for 15: score $\leftarrow$ score + num; 16: end for 17: score $\leftarrow$ score / input;

relationship between individual occurrence is used to determine their similarity. Meanwhile, the inclusion relationship between “occurrence_sum” entries is used to assess the degree of overlap between different occurrences within an “occurrence_sum”. The relevance score for “occurrence_sum” is defined as the proportion of the output “occurrence_sum” that accounts for the input “occurrence_sum”. The calculation formula is provided in Equation (8).

$$\text{Score}(\text{Input}, \text{Output}) = \frac{|\text{Input} \cap \text{Output}|}{|\text{Input}|} \quad (12)$$

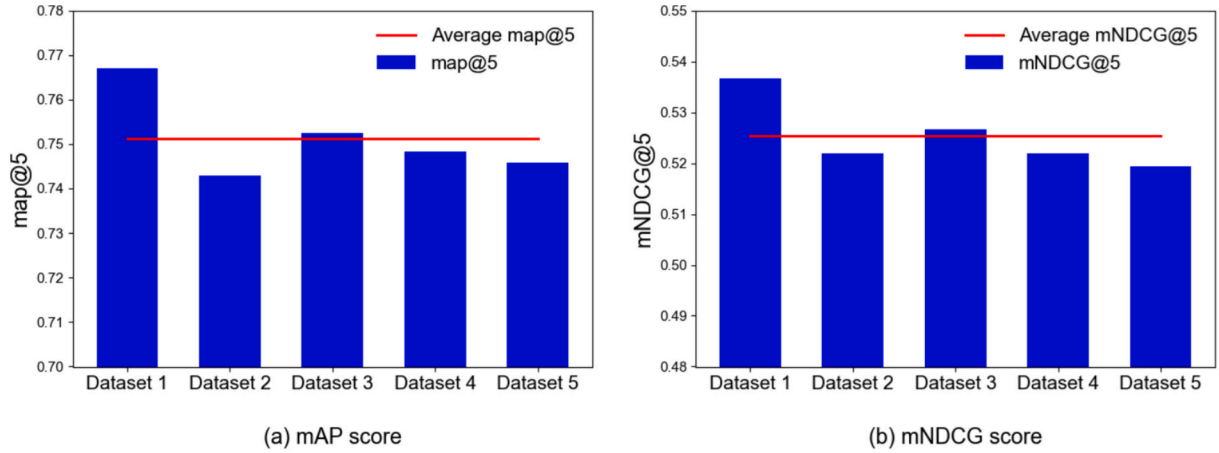
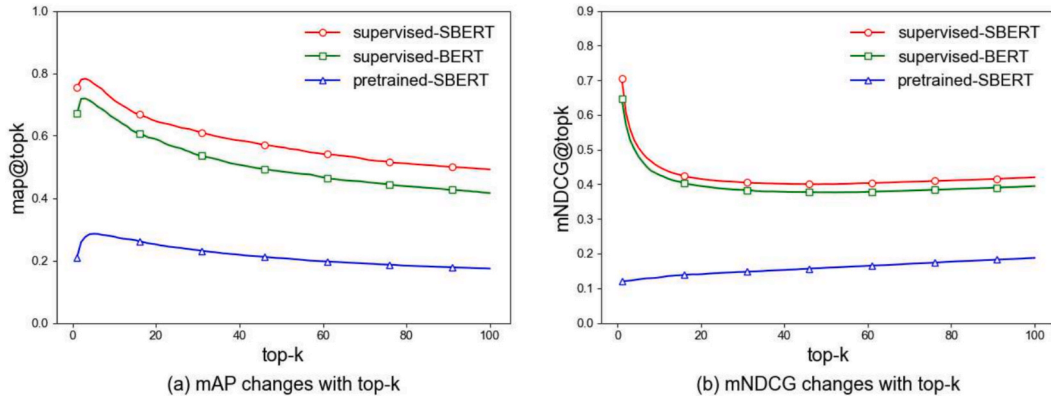
Input is short description “occurrence_sum” of the input, Output is short description “occurrence_sum” corresponding to the output “nar_accf”, and $\text{Input} \cap \text{Output}$ is the occurrence common to both.

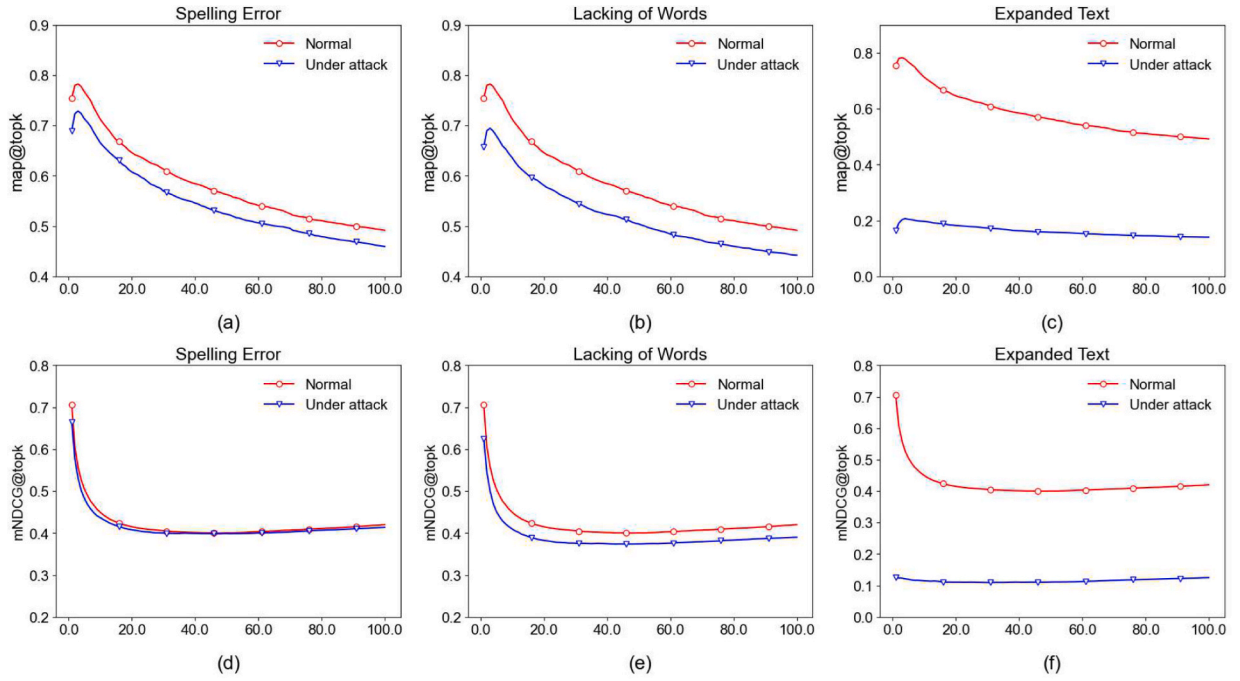
Combining the two inclusion relationships, for any of the inputs and outputs, Table 5 shows the idea of calculating the relevance labels through pseudo-code. $p = \{key_{p,1} = [v_{p,1}^1, v_{p,1}^2], \dots\}$, $o = \{key_{o,1} = [v_{o,1}^1, v_{o,1}^1], \dots\}$, where p denotes the phase of flight and o denotes the type of occurrence, $key_{a,b}$ denotes bth major category of a, $v_{a,b}^c$ th minor category of bth major category of a, and the major category is also an element of minor categories. “The input occurrence_sum is made of $phase_1 + occ_1, \dots, phase_n + occ_n$, which means the equation is $input = [phase_1, occ_1], \dots, [phase_n, occ_n]$ ”; the output occurrence_sum is made of $PHASE_1 + OCC_1, \dots, PHASE_m + OCC_m$, the equation is $output = [PHASE_1, OCC_1], \dots, [PHASE_m, OCC_m]$.

4.1.2. Retriever model fine-tuning and evaluation

The training process uses a five-fold cross-validation approach, with the training set and test set ratio set at 4:1 for each iteration. The data are first randomly shuffled, and the dataset D is then divided into five mutually exclusive subsets of equal size. In each experiment, one subset is used as the test set, while the remaining four subsets are used as the training set, resulting in five experiments in total. This cross-validation method helps eliminate the randomness of dataset division and reduces the likelihood of overfitting to some extent.

The Retriever’s performance is evaluated using two metrics: mean

**Fig. 6.** Cross-validated evaluation scores.**Fig. 7.** Change in evaluation scores with top-k growth.



Notes: Graphs (a), (b) and (c) are mAP@topk scores of models under different attacks. Graphs (d), (e) and (f) are mNDCG@topk scores of models under different attacks.

Fig. 8. Evaluation scores for Retriever under noisy text.

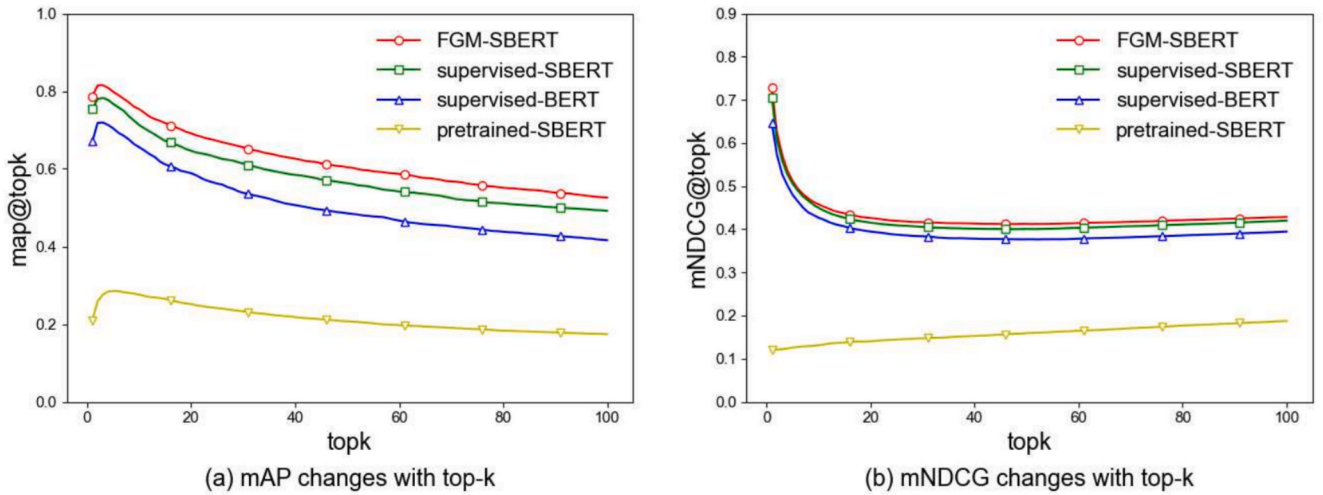


Fig. 9. Variation of evaluation scores with top-k for models.

Average Precision (mAP) and mean Normalized Discounted Cumulative Gain (mNDCG), as shown in Fig. 6 and Fig. 7. The fine-tuned model achieves a maximum mAP score of approximately 0.79 at top-k = 2, while the average score across the cross-validated dataset is around 0.75 at top-k = 5, indicating strong overall performance for the model. The mNDCG score shows a decreasing trend as top-k increases, reaching its maximum value of approximately 0.7 at top-k = 1.

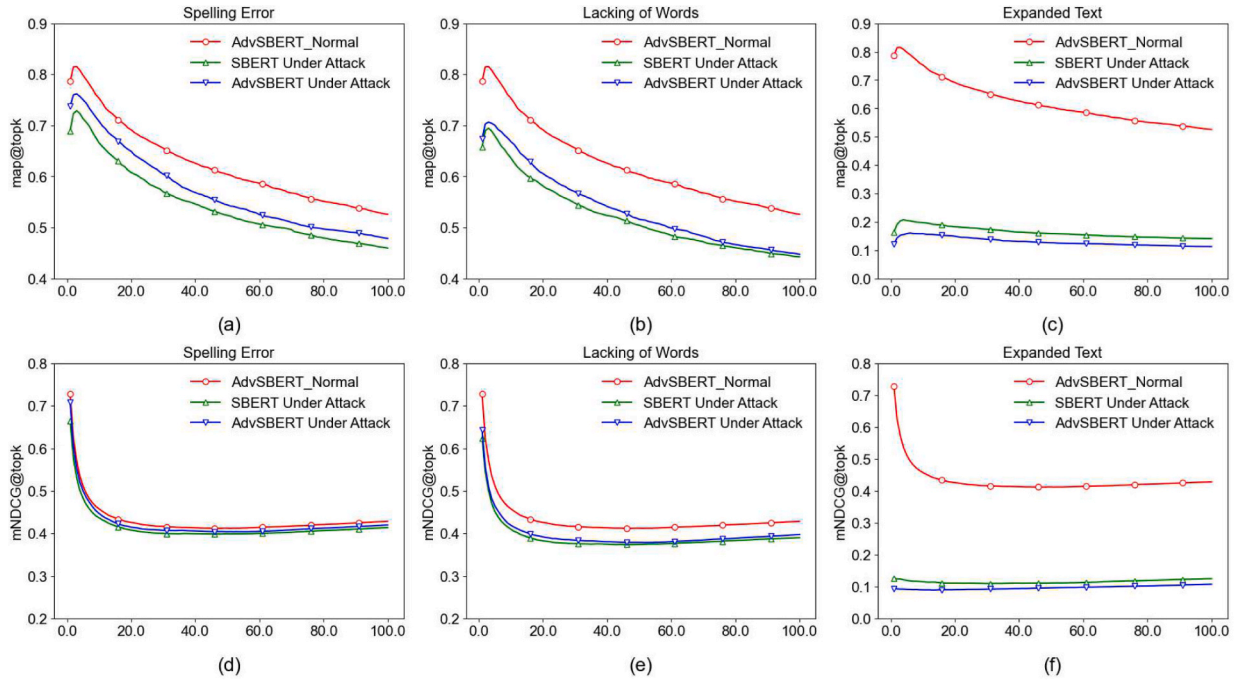
5. Discussion

In this section, we discussed the impact of noise inputs and different prompt designs on RACI's performance. We utilized adversarial training and noise filter to enhance the RACI's ability to resist noise.

5.1. Enhancement of Retriever's anti-noise ability

The accuracy of the Retriever significantly affects the performance of the RACI model. To evaluate the initial Retriever's resilience to noisy information, this paper includes specially designed detection texts for assessment. In the task of assisting with cause localization, Retriever primarily encounters challenges such as fragmented input, missing information, and a low proportion of valid information. To simulate these challenges, the study introduces three types of attack texts: incorrect word text, missing word text, and extended text.

In constructing the perturbed text, each word in the original "nar_accf" field is considered for perturbation with a 10 % probability. For each selected word, perturbation involves either deleting letters,



Notes: Graphs (a), (b) and (c) are $mAP@topk$ scores of models under different attacks. Graphs (d), (e) and (f) are $mNDCG@topk$ scores of models under different attacks.

Fig. 10. AdvSBERT vs SBERT under attack evaluation score comparison.

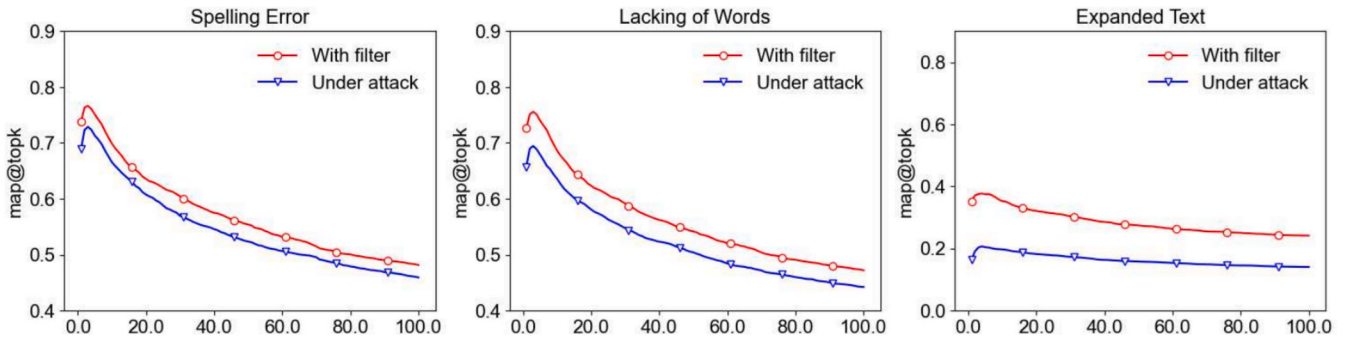


Fig. 11. Filter-aided $mAP@topk$ scoring for SBERT models without sentence segment.

swapping adjacent letters, or adding extra letters with equal probability. To construct the missing-word text, each word in the original “narr_accf” field has a 10 % chance of being selected for perturbation and is then directly deleted. As for the extended text, longer segments of disturbance information are added to the original input. Instead of evaluating individually, a 10 % probability is assigned to each piece of text to determine if it will be extended. Once selected, this text is randomly embedded within a detailed description from an unrelated aviation safety accident event.

The final model was subjected to attacks using the three types of interfering texts, and its performance in terms of mAP and $mNDCG$ scores on the attacked texts was recorded. The performance results of Retriever are shown in Fig. 8.

It is evident that Retriever’s performance declines after being subjected to attacks, with the most significant drop occurring in its ability to recognize expanded text. For example, considering the $mAP@top5$ score, the model’s performance drops by about 5 percent due to spelling errors, approximately 10 percent for missing words, and nearly 55 percent with expanded text. Overall, these disruptions significantly

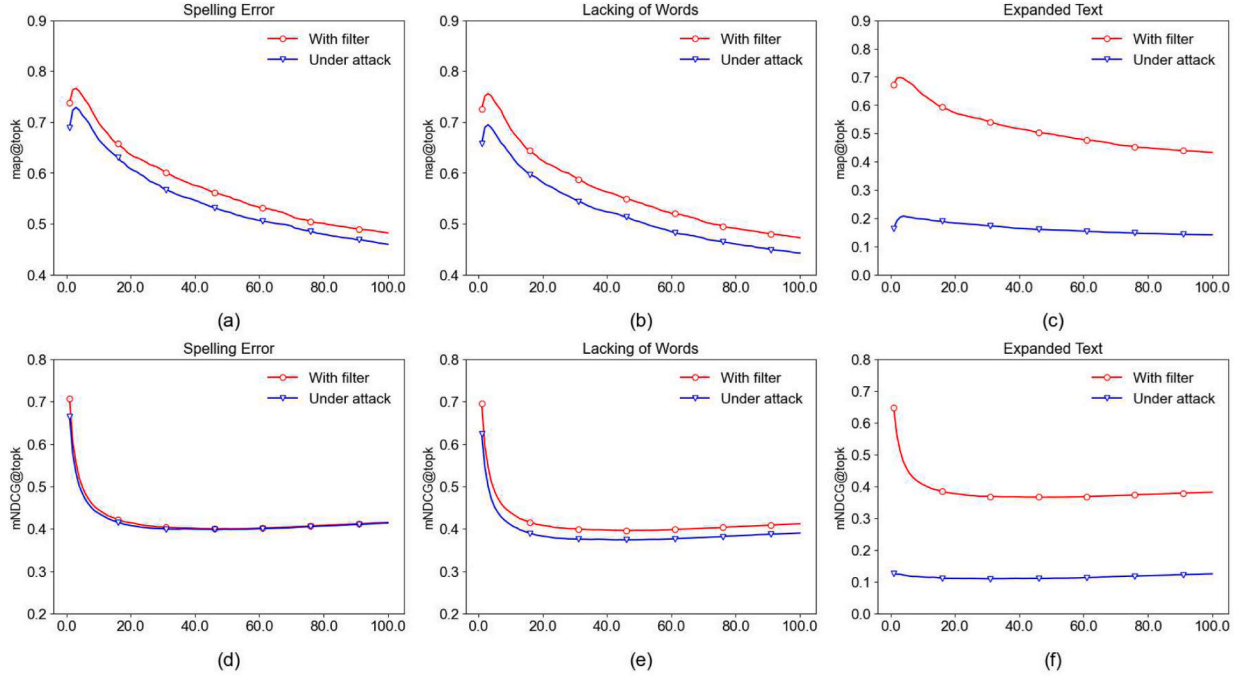
impact the Retriever’s performance.

To strengthen Retriever’s capability in handling disruptive information in real-world applications, enhancements were applied to both the input and model sides. On the model side, adversarial training was implemented to increase robustness. On the input side, filters were introduced to boost performance by reducing the effects of noisy or irrelevant information.

a) Adversarial Training for base model of Retriever.

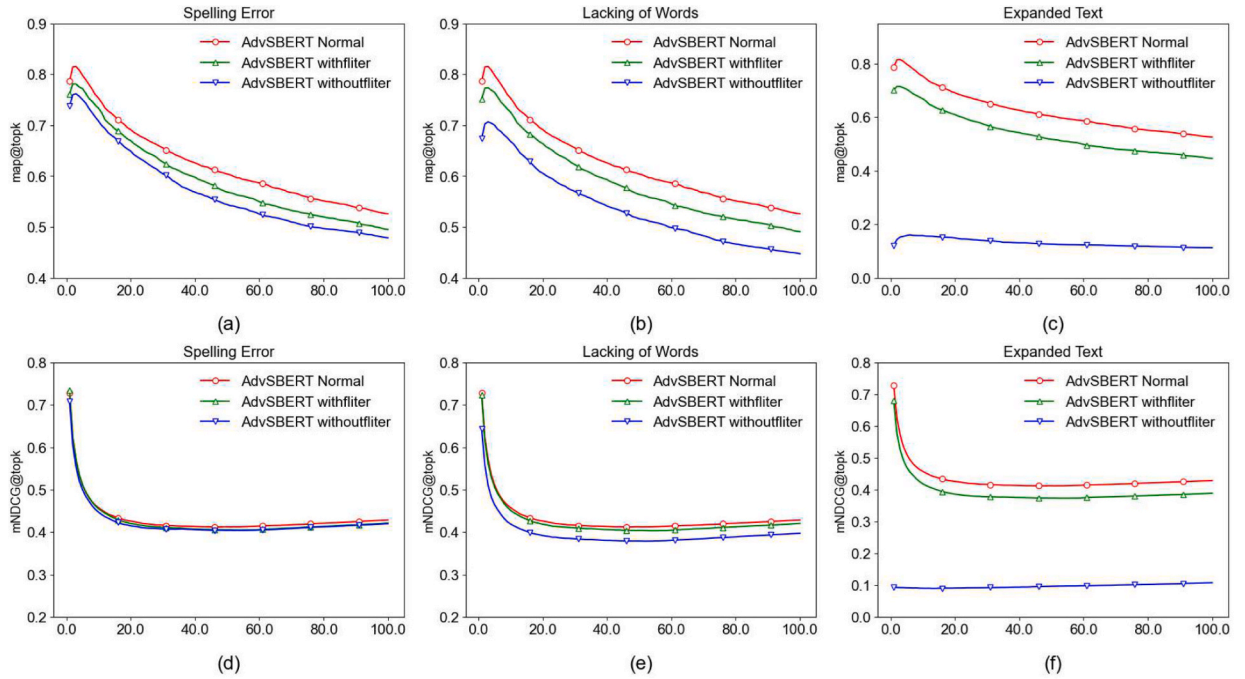
After applying adversarial training, the changes in the Retriever’s evaluation metrics for $mAP@topk$ and $mNDCG@topk$ are illustrated in Fig. 9. It is evident that the model’s performance improves after the addition of adversarial training. The Retriever, enhanced with the SBERT model using the FGM method, outperforms those fine-tuned with the standard SBERT and BERT models.

Additionally, the study further evaluated the robustness of the Retriever with AdvSBERT model against noisy inputs. The model’s performance under three types of attacks—spelling error, lacking of words, and expanded text—without applying any input filtering is presented in Fig. 10.



Notes: Graphs (a), (b) and (c) are $mAP@topk$ scores of models under different attacks. Graphs (d), (e) and (f) are $mNDCG@topk$ scores of models under different attacks.

Fig. 12. Filter-aided scoring for SBERT models with sentence segment.



Notes: Graphs (a), (b) and (c) are $mAP@topk$ scores of models under different attacks. Graphs (d), (e) and (f) are $mNDCG@topk$ scores of models under different attacks.

Fig. 13. Increasing Filter's impact on AdvSBERT's evaluation score.

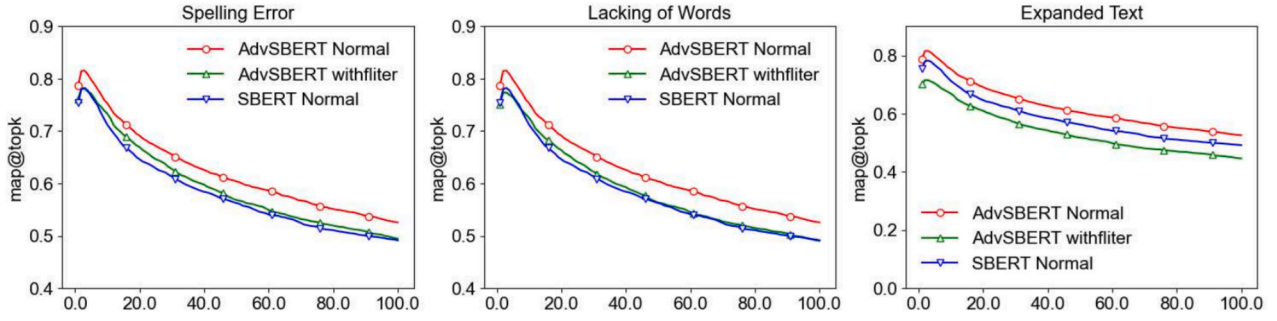


Fig. 14. AdvSBERT vs SBERT's map@topk score comparison.

Table 6

Three types of prompts.

Prompt01	[{'role': 'system', 'content': 'You are an expert in the field of aviation safety.'}, {'role': 'user', 'content': "Question: 'What are the phase of flight and occurrences of the accident?' "}]
Prompt02	[{'role': 'system', 'content': 'You are an expert in the field of aviation safety.'}, {'role': 'user', 'content': "Question: 'What are the common causes of the accidents?'"}]
Prompt03	[{'role': 'system', 'content': ''}, {'role': 'user', 'content': "Question: 'What are the common causes of the accidents?' "}]

Table 7

Comparison of the effects of three types of prompt generation.

Query	climb Airframe/component/system failure/malfunction
Prompt01	1. Beechcraft 1900D Propeller Erosion Shield Accident – Phase: Shortly after take-off. – Occurrences: – The left propeller's erosion shield debonded, ... ### 2. Beechcraft 1900D Aileron Control System Failure – Phase: Immediately after take-off. – Occurrences: – The left down aileron cable broke ... 3. Boeing 737 Engine and Pylon Separation – Phase: Shortly after take-off. – Occurrences: – The engine and pylon separated due to fatigue in the inboard midspar support fitting ... 4. Beechcraft 1900D Horizontal Stabilizer Trim Actuator Malfunction – Phase: During the initial climb after take-off. – Occurrences: – The trim actuator motor was seized, ... 5. Beechcraft 1900D Flight Control Malfunction – Phase: Immediately after take-off. – Occurrences: – The flight crew experienced a hard left roll, requiring full right aileron control ...
Prompt02	1. Propeller Erosion Shield Failure: – Cause: The erosion shield had debonded ... 2. Aileron Cable Failure: – Cause: The left down aileron cable broke, ... 3. Elevator Control System Failure: – Cause: The cable connecting the cockpit to the cabin failed, ... 4. Horizontal Stabilizer Trim Actuator Failure: – Cause: A failure occurred in the horizontal stabilizer trim actuator ... 5. Flight Data Recorder (FDR) Parameter Mismatch: – Cause: ... 6. Horizontal Stabilizer Control System Failure: – Cause: The crew received multiple caution messages on the EICAS,... 7. Wing Outboard Spoiler Deployment: – Cause: A broken outboard left wing spoiler retraction cable led to the spoiler deploying after takeoff....
Prompt03	1. Material Failure: – Propeller Erosion Shield Debonding: ... 2. Mechanical Wear and Fatigue: – Aileron Cable Failure: ... 3. Incorrect Maintenance Practices: – Inadequate Inspection of Propeller Blade Adhesive: ... – Insufficient Inspection of Aileron Control Cables: The inspection procedures lacked detail 4. Procedural Errors: – Overhaul and Repair of Propeller Blade: ... 5. Operator Decisions: – Landing on Snowy Runway: ... 6. Flight Control System Malfunctions: – Horizontal Stabilizer Trim Actuator Seizure: ... – Yaw Damper Malfunction: ...

As shown in Fig. 10, it is clear that the Retriever with AdvSBERT model still experiences significant performance degradation when confronted with noisy inputs. However, it demonstrates better performance

than the SBERT model without adversarial training in scenarios involving spelling errors and missing words. In contrast, in the expanded text attack scenario, the Retriever with AdvSBERT model does not perform as well. Perhaps due to the length of text is too long, the adversarial process of input text is such agnostic that does harm to the training process. This demonstrates that adversarial training alone cannot fully address the issue of noisy inputs, further emphasizing the need for input text filtering.

b) Noise filter.

With the addition of filters at the front-end of the retrieval model, the performance changes under the three types of attacks are shown in Fig. 11. When the TF-IDF method is applied directly to the input text for filtering, its effectiveness in handling expanded text is less satisfactory. This is likely because the excess irrelevant information in the expanded text diminishes the filtering power of the TF-IDF process. To address this problem, the paper segmented the input text by sentence before vectorizing it, ensuring that the information within each vector is more concentrated. The model's performance changes in terms of mAP and mNDCG after filtering are displayed in Fig. 12.

After filtering, the retrieval model's performance significantly improves across all three attack types. The model shows its greatest improvement in resistance to the expanded text attack. For example, in terms of mAP@top5, performance increases by around 3 percent for the spelling error attack, 6 percent for the missing words attack, and an impressive 47 percent for the expanded text attack. After adding filter, the performance of the AdvSBERT model is shown in Fig. 13.

With filtering, AdvSBERT's performance significantly improves, nearing that of a standard model without input perturbations. After adversarial training, the SBERT model with added filters performs under noisy text conditions almost as effectively as the untrained SBERT model does with unaltered inputs. The comparison of mAP@topk evaluation parameters is shown in Fig. 14.

Therefore, this study concludes that, with the integration of adversarial training and filter, the Retriever of RACI effectively defends against input noise.

5.2. Prompt design for aviation accident causal generation

Given that prompt design strongly influences the performance of the generative model, this study compared the effects of three distinct prompts on generation outcomes, as illustrated in Table 6. Under all three prompts, the generative model all produced results that included the causal information. However, the results place different emphasis on the causal information. As shown in Table 7, in prompt 01, the model generated results that included inferences about both the flight phase and occurrence, along with the cause of the accident. This broader context provided additional information about the incident, but the focus was less directed on the causal information itself. In contrast, the second (prompt02) and third (prompt03) prompts placed a stronger emphasis on the accident's cause, generating more concise and focused descriptions. Both prompt02 and prompt03 concentrated on identifying

Table 8
Comparison of the effects of queries.

Query01	Landing – roll Loss of control – on ground/water
Prompt02	The common causes of the accidents described include: Mechanical Issues: A no-brake condition during landing, resulting from using the yellow hydraulic system instead of the green system due to a previous issue. This indicates potential maintenance failures or inadequate system checks. Navigation and Weather Information Discrepancies: The flight crew received outdated information about runway conditions (mu readings), leading to unexpected snow-covered surfaces. This highlights the importance of accurate and timely weather updates. Operational Errors: The use of reverse pitch to slow down the aircraft resulted in the elevator pitch uncoupling mechanism separating, impacting control and requiring immediate crew coordination. This emphasizes the need for proper training and procedures during critical phases of flight. Weather Conditions: Turbulence during descent caused by cloud activity led to an injury among flight attendants. This underscores the importance of accurate weather forecasting and crew preparedness for potential adverse conditions. Control System Failure: The failure of the autopilot to correct the deviation after touchdown on the wet runway could indicate issues with the control systems or pilot intervention timing. Maintenance and Inspection: The engines running at higher idle speeds than recommended suggests potential maintenance issues or operational errors regarding engine performance checks. Each of these factors contributes to a complex interplay between human error, mechanical failures, and environmental conditions, highlighting the multifaceted nature of aviation safety.
Query02	Landing – flare/touch down on ground/water encounter with weather, Landing – roll Loss of control – on ground/water, Landing – roll on ground/water collision with terrain/water
Prompt02	Common causes of accidents in aviation, based on the provided contexts, include: Weather Conditions: Heavy snowfall, reduced visibility due to snow showers, and high winds can impair pilot judgment and aircraft control. Runway Conditions: Snow cover, poor marking, and soft surfaces like snow can lead to loss of control, especially during landing, affecting the aircraft's ability to maintain directional stability. Equipment Failure: Malfunction or failure of critical systems such as engines, landing gears, and steering mechanisms can result in accidents. Human Error: Pilot errors, including misjudgment of speed, incorrect use of controls (like power steering), and failure to follow standard operating procedures can lead to accidents. Maintenance Issues: Faulty maintenance, such as improperly installed or serviced equipment, can compromise safety. Navigation Errors: Misinterpretation of navigation aids or instruments, leading to incorrect approaches or positioning. Limited Training or Experience: Inadequate training or experience can affect a pilot's ability to handle unexpected situations safely. These factors highlight the complex interplay between environmental conditions, human actions, and technical aspects in aviation safety.
Query03	Landing – roll Loss of control – on ground/water, Climb Airframe/component/system failure/malfunction, Descent – normal Altitude deviation, uncontrolled
Prompt02	The common causes of the accident include: Hydraulic System Failure: The loss of the left hydraulic system and subsequent anticipated loss of the right, leading to critical control issues.
Query03	Landing – roll Loss of control – on ground/water, Climb Airframe/component/system failure/malfunction, Descent – normal Altitude deviation, uncontrolled
Prompt02	The common causes of the accident include: Electrical Damage: Lightning strike causing damage to elevator pressure and return hydraulic lines, leading to depletion of hydraulic fluid. Runway Conditions: Landing on a wet runway exacerbated by tire blowouts, contributing to the aircraft's veering off the runway. Weather Issues: Flying into severe thunderstorms without proper weather advisories, including lack of convective SIGMETs and radar training for the crew. Dispatch Oversight: Failure to provide the flight crew with available weather watches, potentially affecting their decision-making during the flight. Maintenance Deficiencies: Discovery of a loose cannon plug preventing the operation of the alternate antiskid system, and inadequate bonding for electrical currents. Design Limitations: The airplane was not designed for substantial amounts of electrical current, affecting its structural integrity under extreme conditions. Flight Crew Experience: The crew's experience with hard landings and vibrations on previous flights may have influenced their actions during the current incident. Equipment Failure: The separation of the left main landing gear tee bolt, leading to its collapse, was found to be related to a pre-existing crack. Maintenance Issues: The landing gear outer cylinder trunnion failure, which had occurred previously, was attributed to fatigue fracture and stress corrosion due to manufacturing defects.
Query04	During rollout the airplane veered immediately right when the power levers were brought into beta. The captain corrected with rudder and braking but was unable to maintain directional control. The aircraft ran off the right side of the runway and collided with a ditch and an embankment. After deplaning, the crew found that the winds were a 60-degree tailwind on runway 12. The windsock's pivot point on the pole was rusted and would not rotate. The aircraft with the same landing weight and a 60-degree 10-knot tailwind would require a 1,600-foot landing roll on the 2,350-foot runway without the assistance of both props in beta.
Prompt02	The common causes of the accidents described include electrical system failures, specifically issues with the power conversion distribution unit leading to a complete electrical power failure, and subsequent cabin smoke and odors possibly related to electrical burning. Another cause is mechanical failure, evident in the damaged nose wheel tire due to a failed tow bar and the tire rim failure, which led to a fatal injury. Additionally, there is an example of a near-collision with wake turbulence from another aircraft, indicating potential navigation errors or misjudgments in flight paths. Lastly, there are issues with emergency procedures and equipment, such as the lack of proper training for flight attendants on operating external air conditioning carts and the absence of clear instructions regarding door openings necessary for cabin pressurization. These examples highlight the importance of robust maintenance practices, clear communication among crew members, and comprehensive safety protocols to prevent accidents.

the causal factors, and the observed difference in generation quality between them was negligible. This implies that either prompt could be utilized effectively when precision is a primary consideration.

In practical operational scenarios, particularly for aviation safety professionals such as cabin crews and safety engineers, prompt01 may offer valuable context by providing more comprehensive descriptions that include accident phases and occurrences. However, this broader approach may also result in less intuitive support for identifying the primary cause of the accident. On the other hand, prompt02 and prompt03, with their more targeted focus on the causal factors, provide more direct and actionable insights, which could be more beneficial in real-time decision-making scenarios where concise and unambiguous causal identification is critical.

Given the negligible difference in performance between prompt02 and prompt03, this study recommends using prompt02 as the final

choice for operational settings. Prompt02 strikes a balance between focusing on accident causes while maintaining clarity and conciseness, which are crucial for rapid understanding and response. This prompt holds significant value in contexts requiring prompt, precise, and actionable causal insights to inform timely safety interventions and support critical decision-making processes.

The study also compares the impact of different input texts on the generated output, with the specific content of the generation shown in Table 8. Query01 represents the case where certain accident-related keywords (typically the most easily observable representations) are used as input. In this scenario, the RACI model demonstrated excellent generative capabilities. The reasons output by the model (e.g., mechanical issues, weather conditions, control system failure, etc.) were well linked to the input keywords. Query02 involved inputting a large number of accident-related keywords, and the potential causes

Table 9

Comparison of the generation outputs between RACI and GPT-4o.

Query	Landing – roll Loss of control – on ground/water
RACI	The common causes of the accidents described include: Mechanical Issues: A no-brake condition during landing, resulting from using the yellow hydraulic system instead of the green system due to a previous issue. This indicates potential maintenance failures or inadequate system checks. Navigation and Weather Information Discrepancies: The flight crew received outdated information about runway conditions (mu readings), leading to unexpected snow-covered surfaces. This highlights the importance of accurate and timely weather updates. Operational Errors: The use of reverse pitch to slow down the aircraft resulted in the elevator pitch uncoupling mechanism separating, impacting control and requiring immediate crew coordination. This emphasizes the need for proper training and procedures during critical phases of flight. Weather Conditions: Turbulence during descent caused by cloud activity led to an injury among flight attendants. This underscores the importance of accurate weather forecasting and crew preparedness for potential adverse conditions. Control System Failure: The failure of the autopilot to correct the deviation after touchdown on the wet runway could indicate issues with the control systems or pilot intervention timing. Maintenance and Inspection: The engines running at higher idle speeds than recommended suggests potential maintenance issues or operational errors regarding engine performance checks. Each of these factors contributes to a complex interplay between human error, mechanical failures, and environmental conditions, highlighting the multifaceted nature of aviation safety.
GPT-4o	Weather Conditions: Wind shear or crosswinds during landing can lead to loss of control, especially if the aircraft fails to maintain proper alignment with the runway. Slippery runway conditions, such as rain, ice, or snow, can make it difficult for the aircraft to decelerate properly, increasing the risk of skidding or loss of control. Pilot Error: Incorrect control inputs during landing, such as improper use of throttle, a misjudged flare, or incorrect rudder or aileron inputs, can lead to roll instability. Failure to compensate for wind conditions, such as wind gusts or crosswinds, especially if the aircraft is not aligned with the runway. Aircraft Mechanical Failure: Landing gear failure (e.g., failure of one or more landing gears) can cause an aircraft to veer off course or experience a loss of control after touchdown. Flap or aileron malfunctions can reduce control authority during landing, making the aircraft difficult to stabilize on the ground. Runway Conditions: Runway surface damage such as cracks or debris can affect the stability of the aircraft after landing, especially during a roll-out. Improper runway design or poor maintenance leading to an uneven surface or a runway with inadequate length for stopping or maneuvering. Flight Path Misalignment: Incorrect approach angle or failure to perform a proper landing approach can result in an unstable descent that leads to roll instability or veering off the centerline. Environmental Factors: Bird strikes during landing or shortly after touch-down may cause control loss or damage to the aircraft. Obstructions on the runway (such as vehicles, wildlife, or equipment) may cause pilots to react unexpectedly, leading to a loss of control. Aircraft Weight and Balance Issues: Improper load distribution or exceeding weight limits can lead to instability during the landing roll, particularly if the aircraft is tail-heavy or unbalanced.

identified by the RACI model (e.g., Weather Conditions, Equipment Failure, Runway Conditions, etc.) remained reasonable and closely related to the input keywords. This indicates that the retrieval module of the RACI model performs as expected in scenarios with highly relevant keywords. Query03 also involved a large number of keywords as input, but irrelevant keywords were used. It was observed that the potential causes identified by RACI were still somewhat related to the keywords. However, because the retrieval unit matched with unrelated accidents, the causes identified by RACI could belong to completely different types of accidents, limiting their reference value. Query04 directly used non-keyword textual information as input. In this case, although the RACI model's generated output included mechanical failure, the full text description revealed that the model mistakenly identified the accident type. A possible explanation for this phenomenon is that the filtering

Table 10

Comparison of the generation of raw text and noisy text in the same prompt.

Query	climb Airframe/component/system failure/malfunction
Normal	- Propeller Erosion Shield Failure: – Cause: The erosion shield had debonded ... - Aileron Cable Failure: – Cause: The left down aileron cable broke, ... - Elevator Control System Failure: – Cause: The cable connecting the cockpit to the cabin failed, ... - Horizontal Stabilizer Trim Actuator Failure: – Cause: A failure occurred in the horizontal stabilizer trim actuator ... - Flight Data Recorder (FDR) Parameter Mismatch: – Cause: ... - Horizontal Stabilizer Control System Failure: – Cause: The crew received multiple caution messages on the Electronic Information Control Advisory System (EICAS),... - Wing Outboard Spoiler Deployment: – Cause: A broken outboard left wing spoiler retraction cable led to the spoiler deploying after takeoff....
Spelling error	- Propeller Erosion Shield Failure: – Cause: Debonding of the erosion shield leading to structural integrity issues. ... – Impact: ... - Aileron Control System Failures: – Cause: Issues ranged from a broken left down aileron cable, ... – Impact: ... - Horizontal Stabilizer Trim Actuator Malfunction: – Cause: A seized actuator motor prevented the trim system from functioning correctly, necessitating manual trimming with excessive force. – Impact: ... - Engine and Pylon Failure: – Cause: Fatigue cracks in the pylon inboard midspar support fitting caused the engine and pylon to separate from the aircraft. – Impact: ... - Flight Control System Issues: – Cause: Problems included faulty cables, malfunctioning stabilizer trim systems, ... – Impact: ... - Maintenance and Inspection Shortcomings: – Cause: Lack of proper inspection procedures... – Impact: ... - Operator Practices: – Cause: Reduced inspection intervals for cables and failure to follow recommended maintenance procedures. – Impact: ...]
Expanded text	- Propeller Erosion Shield Failure: – Cause: Propeller erosion shields debonding due to adhesive issues, ... - Aileron Control System Issues: – Cause: Fracture in the aileron control system cables, possibly due to fatigue and wear, ... - Horizontal Stabilizer Trim Actuator Failure: – Cause: A failure in the trim actuator motor that prevents the stabilization of the aircraft during flight. ... - Engine and Pylon Separation: – Cause: Fatigue cracking in the pylon inboard midspar support fitting, ... - Flight Data Recorder (FDR) Parameter Inconsistencies: – Cause: Discrepancies in FDR readings related to roll control parameters, ... - Flight Control Malfunction Post-Takeoff: – Cause: A combination of factors including yaw and roll control issues, ... - Wheel System Failure: – Cause: Failure of the wheel bearings due to inadequate lubrication or pre-existing damage, ...]
Lacking of words	- Propeller Erosion Shield Failure: The debonding and subsequent penetration of the passenger cabin ... – Material Fatigue: Over time, the propeller blade and shield may experience fatigue, especially if subjected to vibrations during flight, leading to disbonding. – Contamination: ... - Aileron Control System Failures: – Worn Aileron Cables: The failure of the left down aileron cable ... – Incorrect Cable Replacement Interval: ... - Horizontal Stabilizer Trim Actuator Failure: – Seized Actuator Motor: The failure of the motor, indicated by the inability to move the stabilizer through manual trimming ... - Flight Data Recorder (FDR) Issues: – Incorrect Calibration or Maintenance ... - Engine Pylon Failure: – Fatigue Crack Propagation ... - Spoilers Deployment: – Worn Retraction Cable: ...

unit in the RACI model has been specifically optimized for keyword-based processing. Consequently, its effectiveness may decrease when applied to full-text inputs that contain a high proportion of irrelevant information, leading to performance degradation.

To validate the performance of the RACI model in extracting causes of aviation accidents, we compared the generative results of the RACI model with those of the widely used GPT-4o model. The results

generated from the same set of keywords are shown in Table 9. From the generated results, it can be observed that both the RACI model and GPT-4o generated causes closely related to the input keywords. However, a key difference is that the RACI model, in association with real-world accidents, extracted valuable information, such as operational behaviors, that were closely linked to the causes from historical incidents. In contrast, the output from GPT-4o was more general. For the decision-making process, the output from the RACI model, which better locates the causes, is more suitable.

As shown in Table 10, under the same prompt, the probable causes of accidents summarized by the model are essentially the same when using noisy text or original text as input, demonstrating the model's resilience to noisy input.

Combined with Retriever and Generator, the RACI model can provide actionable insights and decision-making support without requiring deep knowledge of data science or machine learning. RACI model automates the extraction of causals from complex aviation accident reports, converting unstructured data into interpretable insights.

The practical implications of RACI model include:

1. Integration into pilot training and aviation safety workflows: The RACI model identifies causal factors from historical accident data. This information can be used to create scenario-based training modules. These modules help pilots improve decision-making in high-risk situations by simulating real-world scenarios, such as accidents caused by specific weather conditions or system failures.
2. Support for accident investigation: The model's retrieval module provides investigators with quick access to similar historical accidents. This capability allows investigators to focus on key risk propagation paths, shortening the time required to identify root causes and recommend preventive measures.
3. Real-time decision-making assistance: The generation module provides real-time insights by retrieving relevant historical accidents. This supports operational teams in making informed decisions under time constraints, such as during in-flight emergencies.

6. Conclusion

This paper proposed a novel Retrieval-Augmented Generation (RAG)-aided model to assist in identifying the causes of aviation safety incidents. The model consists of a retrieval module and a generation module. The RACI model reveals significant potential for scalability across domains beyond aviation. This scalability is largely due to its modular architecture. The RACI model is built upon a retrieval-augmented framework that integrates domain-specific knowledge and historical data. By replacing the aviation accident database with domain-specific datasets, the model can be adapted to other industries. For example, in healthcare, electronic health records or incident reports from hospitals can serve as the knowledge base for identifying causal factors in medical errors or adverse events. The generation module, which uses a pre-trained large language model (e.g., Qwen2-7b), can be adapted to various contexts through carefully designed prompts tailored to the domain.

While the RACI model can be scaled, there are still some limitations for the present RACI model. Not like the analytical model approved by Murtza et al. (2024). In RACI model, domain-specific datasets and expertise are required to refine the retrieval and generation modules. The performance of retrieval module heavily relies on the quality of training dataset. The noise resistance mechanisms also need adjustments to accommodate variations in data quality across domains, especially the filter. In NTSB dataset, the "occurrence sum" can save as the reference for the noise filtering module. The NTSB dataset also has an adequate labeling system for the training process. Other aviation datasets, particularly those from international organizations or non-standardized sources, could lack similar structured fields, making it difficult to replicate the same level of noise filtering and data

preprocessing.

The RACI model currently can only accept keywords as input, and its ability to process complete accident file descriptions is inadequate. For texts that are extremely vague and contain excessive irrelevant information, the filtering unit cannot achieve sufficient improvement, and the model's processing capability for similar inputs remains limited. The model currently only has the ability to retrieve potential causal factors from historical accidents and does not yet have the capability to generate solutions, which limits its potential in practical application scenarios. The output results generated by the model lack sufficient conciseness, which increases the cognitive stress on users and hinder efficient information comprehension.

Future work can focus on the following directions. First, improvement of the prompt engineering for the current RACI model is essential, as better prompt design can make its outputs more precise and refined, enabling quicker transmission of accurate causal information to cabin crew. Secondly, supplementing the accident database with solution methods, This will help RACI model get the ability to give possible solutions when encounters with an accident. Thirdly, further enhancement of the filter. The current filter limits the format of input, potentially reducing the original information and affecting model performance. Developing new methods to handle noise would be a practical area for further investigation. Finally, training RACI model on other datasets. Considering RACI's potential for scalability across domains beyond aviation, it is necessary to enhance the model using datasets from other fields.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Abesamis, P. P. R., de Dios Bulos, R., & Ching, M. (2020, October). Improving aviation incidents using association rule mining algorithm and time series analysis. In IOP Conference Series: Materials Science and Engineering (Vol. 946, No. 1, p. 012005). IOP Publishing.
- Aviator. (2017). IATA: 2036 Forecast Reveals Air Passengers Will Nearly Double to 7.8 Billion. Retrieved from <https://newsroom.aviator.aero/2036-forecast-reveals-air-passengers-will-nearly-double-to-7-8-billion/>. Accessed October 17, 2024.
- Beyer, P. (2015). Proposal for a STS Evaluation Framework for STS based Applications (Doctoral dissertation, Master's thesis, Technische Universität Darmstadt, March. Available online: <https://www.ukp.tu-darmstadt.de/publications/details>). Beyer, Philip. Proposal for a STS Evaluation Framework for STS based Applications. Diss. Master's thesis, Technische Universität Darmstadt, March. Available online: <https://www.ukp.tu-darmstadt.de/publications/details>, 2015.
- Bharathi, A., Ramdin, R., Babu, P., Menon, V. K., Jayaramakrishnan, C., & Lakshmikummar, S. (2024). A hybrid named entity recognition system for aviation text. EAI Endorsed Transactions on Scalable Information Systems, 11(1). Bharathi, A., et al. "A hybrid named entity recognition system for aviation text." EAI Endorsed Transactions on Scalable Information Systems 11.1 (2024).
- Chang, H., Li, L., Huang, J., Zhang, Q., & Chin, K. S. (2022). Tracking traffic congestion and accidents using social media data: A case study of Shanghai. Accident Analysis & Prevention, 169, 106618. Chang, Haoliang, et al. "Tracking traffic congestion and accidents using social media data: A case study of Shanghai." Accident Analysis & Prevention 169 (2022): 106618.
- Chen, L., Xu, J., Wu, T., & Liu, J. (2024). Information Extraction of Aviation Accident Causation Knowledge Graph: An LLM-Based Approach. Electronics, 13(19), 3936. Chen, Lu, et al. "Information Extraction of Aviation Accident Causation Knowledge Graph: An LLM-Based Approach." Electronics 13.19 (2024): 3936.
- Chen, Z., Huang, K., Wu, L., Zhong, Z., & Jiao, Z. (2022). Relational graph convolutional network for text-mining-based accident causal classification. Applied Sciences, 12 (5), 2482. Chen, Zaili, et al. "Relational graph convolutional network for text-mining-based accident causal classification." Applied Sciences 12.5 (2022): 2482.
- Cuconasu, F., Trappolini, G., Siciliano, F., Filice, S., Campagnano, C., Maarek, Y., ... & Silvestri, F. (2024, July). The power of noise: Redefining retrieval for rag systems. In Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (pp. 719-729). Cuconasu, Florin, et al. "The

- power of noise: Redefining retrieval for rag systems." Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval. 2024.
- de Zarzà, I., de Curtò, J., Roig, G., & Calafate, C. T. (2023). LLM multimodal traffic accident forecasting. *Sensors*, 23(22), 9225. de Zarzà, Irene, et al. "LLM multimodal traffic accident forecasting." *Sensors* 23.22 (2023): 9225.
- Gao, Y., Zhu, G., Duan, Y., & Mao, J. (2024). Semantic Encoding Algorithm for Classification and Retrieval of Aviation Safety Reports. *IEEE Transactions on Automation Science and Engineering*. Gao, Yubing, et al. "Semantic Encoding Algorithm for Classification and Retrieval of Aviation Safety Reports." *IEEE Transactions on Automation Science and Engineering* (2024).
- Gou, Q., Dong, Y., Wu, Y., & Ke, Q. (2024). RRGcode: Deep hierarchical search-based code generation. *Journal of Systems and Software*, 211, 111982. Gou, Qianwen, et al. "RRGcode: Deep hierarchical search-based code generation." *Journal of Systems and Software* 211 (2024): 111982.
- Guu, K., Lee, K., Tung, Z., Pasupat, P., & Chang, M. (2020, November). Retrieval augmented language model pre-training. In *International conference on machine learning* (pp. 3929-3938). PMLR. Guu, Kelvin, et al. "Retrieval augmented language model pre-training." *International conference on machine learning*. PMLR, 2020.
- ICAO. (2022) ICAO 2022 Safety Report. Retrieved from https://www.icao.int/safety/Documents/ICAO_SR_2022.pdf. Accessed October 17, 2024. ICAO Safety F. The annual ICAO safety report [EB/OL]. [2022]. https://www.icao.int/safety/Documents/ICAO_SR_2022.pdf.
- Jaradat, S., Nayak, R., Paz, A., Ashqar, H. I., & Elhenawy, M. (2024). Multitask learning for crash analysis: A Fine-Tuned LLM framework using twitter data. *Smart Cities*, 7 (5), 2422-2465. Jaradat, Shadi, et al. "Multitask Learning for Crash Analysis: A Fine-Tuned LLM Framework Using Twitter Data." *Smart Cities* 7.5 (2024): 2422-2465.
- Jiang, B., & Wang, K. (2024). Railway accident causation prediction with improved transformer model based on lexical information and contextual relationships. *Knowledge-Based Systems*, 296, 111897. Jiang, Bin, and Keming Wang. "Railway accident causation prediction with improved transformer model based on lexical information and contextual relationships." *Knowledge-Based Systems* 296 (2024): 111897.
- Jiang, Z., Xu, F. F., Gao, L., Sun, Z., Liu, Q., Dwivedi-Yu, J., ... & Neubig, G. (2023). Active retrieval augmented generation. *arXiv preprint arXiv:2305.06983*. Jiang, Zhengbao, et al. "Active retrieval augmented generation." *arXiv preprint arXiv:2305.06983* (2023).
- Kiciman, E., Ness, R., Sharma, A., & Tan, C. (2023). Causal reasoning and large language models: Opening a new frontier for causality. *arXiv preprint arXiv:2305.00050*. Kiciman, Emre, et al. "Causal reasoning and large language models: Opening a new frontier for causality." *arXiv preprint arXiv:2305.00050* (2023).
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474. Lewis, Patrick, et al. "Retrieval-augmented generation for knowledge-intensive nlp tasks." *Advances in Neural Information Processing Systems* 33 (2020): 9459-9474.
- Liu, C., & Yang, S. (2022). Using text mining to establish knowledge graph from accident/incident reports in risk assessment. *Expert Systems with Applications*, 207, Article 117991.
- Liu, Y., Jiang, C., & Zhao, H. (2018). Using contextual features and multi-view ensemble learning in product defect identification from online discussion forums. *Decision Support Systems*, 105, 1-12. Liu, Yao, Cuiqing Jiang, and Huimin Zhao. "Using contextual features and multi-view ensemble learning in product defect identification from online discussion forums." *Decision Support Systems* 105 (2018): 1-12.
- Marcus, G. (2020). The next decade in AI: four steps towards robust artificial intelligence. *arXiv preprint arXiv:2002.06177*. Marcus, Gary. "The next decade in AI: four steps towards robust artificial intelligence." *arXiv preprint arXiv:2002.06177* (2020).
- Min, B., Ross, H., Sulem, E., Veyseh, A. P. B., Nguyen, T. H., Sainz, O., ... & Roth, D. (2023). Recent advances in natural language processing via large pre-trained language models: A survey. *ACM Computing Surveys*, 56(2), 1-40. Min, Bonan, et al. "Recent advances in natural language processing via large pre-trained language models: A survey." *ACM Computing Surveys* 56.2 (2023): 1-40.
- Min, S., Lee, K., Chang, M. W., Toutanova, K., & Hajishirzi, H. (2021). Joint passage ranking for diverse multi-answer retrieval. *arXiv preprint arXiv:2104.08445*. Min, Sewon, et al. "Joint passage ranking for diverse multi-answer retrieval." *arXiv preprint arXiv:2104.08445* (2021).
- Miyato, T., Dai, A. M., & Goodfellow, I. (2016). Adversarial training methods for semi-supervised text classification. *arXiv preprint arXiv:1605.07725*. Miyato, Takeru, Andrew M. Dai, and Ian Goodfellow. "Adversarial training methods for semi-supervised text classification." *arXiv preprint arXiv:1605.07725* (2016).
- Wu, Z., & Ma, G. (2024). NLP-based approach for automated safety requirements information retrieval from project documents. *Expert Systems with Applications*, 239, Article 122401.
- Nazeri, Z., & Bloedorn, E. (2004). Exploiting available domain knowledge to improve mining aviation safety and network security data. The MITRE Corporation, McLean, Virginia, 22102. Nazeri, Zohreh, and Eric Bloedorn. "Exploiting available domain knowledge to improve mining aviation safety and network security data." The MITRE Corporation, McLean, Virginia 22102 (2004).
- Ng, C. B. R., Bil, C., Sardina, S., & O'bre, T. (2022). Designing an expert system to support aviation occurrence investigations. *Expert Systems with Applications*, 207, 117994.
- Ni, Z. J., & Zhang, Q. (2023). Mining the Critical Factors of Hazardous Material Road Transportation Accidents From Chinese Investigation Reports. *IEEE Transactions on Intelligent Transportation Systems*, 24(10), 11121-11131. Ni, Zi-Jian, and Qi Zhang. "Mining the Critical Factors of Hazardous Material Road Transportation Accidents From Chinese Investigation Reports." *IEEE Transactions on Intelligent Transportation Systems* 24.10 (2023): 11121-11131.
- Parvez, M. R., Ahmad, W. U., Chakraborty, S., Ray, B., & Chang, K. W. (2021). Retrieval augmented code generation and summarization. *arXiv preprint arXiv:2108.11601*. Parvez, Md Rizwan, et al. "Retrieval augmented code generation and summarization." *arXiv preprint arXiv:2108.11601* (2021).
- Peng, B., Galley, M., He, P., Cheng, H., Xie, Y., Hu, Y., ... & Gao, J. (2023). Check your facts and try again: Improving large language models with external knowledge and automated feedback. *arXiv preprint arXiv:2302.12813*. Peng, Baolin, et al. "Check your facts and try again: Improving large language models with external knowledge and automated feedback." *arXiv preprint arXiv:2302.12813* (2023).
- Perboli, G., Gajetti, M., Fedorov, S., & Giudice, S. L. (2021). Natural Language Processing for the identification of Human factors in aviation accidents causes: An application to the SHEL methodology. *Expert Systems with Applications*, 186, 115694. Perboli, Guido, et al. "Natural Language Processing for the identification of Human factors in aviation accidents causes: An application to the SHEL methodology." *Expert Systems with Applications* 186 (2021): 115694.
- Ram, O., Levine, Y., Dalmedigos, I., Muhlga, D., Shashua, A., Leyton-Brown, K., & Shoham, Y. (2023). In-context retrieval-augmented language models. *Transactions of the Association for Computational Linguistics*, 11, 1316-1331. Ram, Ori, et al. "In-context retrieval-augmented language models." *Transactions of the Association for Computational Linguistics* 11 (2023): 1316-1331.
- Murtza, I., Kim, J. Y., & Adnan, M. (2024). Predicting the Performance of Ensemble Classification Using Conditional Joint Probability.
- Reimers, N. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *arXiv preprint arXiv:1908.10084*. Reimers, N. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks." *arXiv preprint arXiv:1908.10084* (2019).
- Reimers, N., Beyer, P., & Gurevych, I. (2016, December). Task-oriented intrinsic evaluation of semantic textual similarity. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers* (pp. 87-96). Reimers, Nils, Philip Beyer, and Iryna Gurevych. "Task-oriented intrinsic evaluation of semantic textual similarity." *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*. 2016.
- Samarajeewa, C., De Silva, D., Osipov, E., Alahakoon, D., & Manic, M. (2024). In July). *Causal Reasoning in Large Language Models using Causal Graph Retrieval Augmented Generation* (pp. 1-6). IEEE.
- Shi, C., Su, Y., Yang, C., Yang, Y., & Cai, D. (2023). Specialist or generalist? instruction tuning for specific NLP tasks. *arXiv preprint arXiv:2310.15326*.
- Siriwardhana, S., Weerasekera, R., Wen, E., Kaluarachchi, T., Rana, R., & Nanayakkara, S. (2023). Improving the domain adaptation of retrieval augmented generation (RAG) models for open domain question answering. *Transactions of the Association for Computational Linguistics*, 11, 1-17. Siriwardhana, Shamane, et al. "Improving the domain adaptation of retrieval augmented generation (RAG) models for open domain question answering." *Transactions of the Association for Computational Linguistics* 11 (2023): 1-17.
- Tanguy, L., Tulechki, N., Urieli, A., Hermann, E., & Raynal, C. (2016). Natural language processing for aviation safety reports: From classification to interactive analysis. *Computers in Industry*, 78, 80-95. Tanguy, Ludovic, et al. "Natural language processing for aviation safety reports: From classification to interactive analysis." *Computers in Industry* 78 (2016): 80-95.
- Tikayat Ray, A., Bhat, A. P., White, R. T., Nguyen, V. M., Pinon Fischer, O. J., & Mavris, D. N. (2023). Examining the potential of generative language models for aviation safety analysis: Case study and insights using the aviation safety reporting system (asrs). *Aerospace*, 10(9), 770. Tikayat Ray, Archana, et al. "Examining the potential of generative language models for aviation safety analysis: Case study and insights using the aviation safety reporting system (asrs)." *Aerospace* 10.9 (2023): 770.
- Xu, Y., Ishii, E., Cahyawijaya, S., Liu, Z., Winata, G. I., Madotto, A., ... & Fung, P. (2021). Retrieval-free knowledge-grounded dialogue response generation with adapters. *arXiv preprint arXiv:2105.06232*. Xu, Yan, et al. "Retrieval-free knowledge-grounded dialogue response generation with adapters." *arXiv preprint arXiv:2105.06232* (2021).
- Xu, Z., Cruz, M. J., Guevara, M., Wang, T., Deshpande, M., Wang, X., & Li, Z. (2024, July). Retrieval-augmented generation with knowledge graphs for customer service question answering. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 2905-2909). Xu, Zhentao, et al. "Retrieval-augmented generation with knowledge graphs for customer service question answering." *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2024.
- Yan, K., Wang, Y., Jia, L., Wang, W., Liu, S., & Geng, Y. (2023). A content-aware corpus-based model for analysis of marine accidents. *Accident Analysis & Prevention*, 184, 106991. Yan, Kai, et al. "A content-aware corpus-based model for analysis of marine accidents." *Accident Analysis & Prevention* 184 (2023): 106991.
- Yang, A., Yang, B., Hui, B., Zheng, B., Yu, B., Zhou, C., ... & Huang, F. Qwen2 technical report. *arXiv* 2024. *arXiv preprint arXiv:2407.10671*. Yang, An, et al. "Qwen2 technical report." *arXiv preprint arXiv:2407.10671* (2024).
- Yang, C., & Huang, C. (2023). Natural Language Processing (NLP) in Aviation Safety: Systematic Review of Research and Outlook into the Future. *Aerospace*, 10(7), 600. Yang, Chuyang, and Chenyu Huang. "Natural Language Processing (NLP) in Aviation Safety: Systematic Review of Research and Outlook into the Future." *Aerospace* 10.7 (2023): 600.
- Zhang, X., & Mahadevan, S. (2019). Ensemble machine learning models for aviation incident risk prediction. *Decision Support Systems*, 116, 48-63. Zhang, Xiaoge, and Sankaran Mahadevan. "Ensemble machine learning models for aviation incident risk prediction." *Decision Support Systems* 116 (2019): 48-63.

- Zhang, X., & Mahadevan, S. (2021). Bayesian network modeling of accident investigation reports for aviation safety assessment. *Reliability Engineering & System Safety*, 209, Article 107371.
- Zhao, D., Liu, X., Ning, B., & Liu, C. (2022, April). HRG: A hybrid retrieval and generation model in multi-turn dialogue. In International Conference on Database Systems for Advanced Applications (pp. 181-196). Cham: Springer International Publishing.
- Zhao, Deji, et al. "HRG: A hybrid retrieval and generation model in multi-turn dialogue." International Conference on Database Systems for Advanced Applications. Cham: Springer International Publishing, 2022.
- Zhou, D., Zhuang, X., Cai, J., Zuo, H., Zhao, X., & Xiang, J. (2024). An ensemble model using temporal convolution and dual attention gated recurrent unit to analyze risk of civil aircraft. *Expert Systems with Applications*, 236, Article 121423.
- Zhou, Q., Liu, C., Duan, Y., Sun, K., Li, Y., Kan, H., ... & Hu, J. (2024). GastroBot: a Chinese gastrointestinal disease chatbot based on the retrieval-augmented generation. *Frontiers in Medicine*, 11, 1392555.
- Zhou, Qingqing, et al. "GastroBot: a Chinese gastrointestinal disease chatbot based on the retrieval-augmented generation." *Frontiers in Medicine* 11 (2024): 1392555.